

Dephaze: A Generative Phase-Field Framework for Cosmological and Quantum Anomalies

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Abstract

We present DEPHAZE, an axiomatic generative framework treating observable reality as a projection from a timeless ground state (Ω_0) to manifest configurations (Ψ). The theory derives cosmological and quantum phenomena from five closed axioms without introducing dark matter, dark energy, or wavefunction collapse.

Key results: (1) CMB temperature anisotropy $\Delta T/T \approx 1.60 \times 10^{-4}$ derived parameter-free from primordial power spectrum normalization; (2) Galaxy rotation curves reproduced parameter-free for the **complete SPARC 175-galaxy database** using the DEPHAZE V66 equation $v_{\text{pred}}^2 = v_{\text{obs}}^2 + T_i \cdot S(x_i)$ (median RMS = 6.55 km/s, 114/175 galaxies below 8 km/s threshold, zero free parameters); Leave-One-Out validation confirms median 5.98 km/s; random-shuffle control yields median 24.7 km/s ($3.8\times$ worse), excluding spurious pattern; SPARC input data is Newton-violating by construction—DEPHAZE recovers structure without dark matter or baryonic mass input; (3) Cosmic acceleration from phase-time scaling (ε -correction) without vacuum energy, resolving H_0 and σ_8 tensions; (4) Flyby anomalies predicted via Φ^3 four-point quadrangulation with zero free parameters per mission: 10/10 missions consistent, sign correct in all cases including the Cassini sign reversal that Anderson’s empirical formula cannot reproduce; Pioneer anomaly recovered at 96.8% from 266 digitized Doppler points with zero free parameters; (5) Quantum entanglement as shared Ω_0 -origin; the singlet state $|\Psi^-\rangle$ is derived from time-reversal invariance of Ω_0 via the GNS construction (not postulated); the Tsirelson bound $2\sqrt{2}$ is identified as a *virtual asymptote*—saturated only at $\det(\rho) = 0$, which requires $T = 0$ K and is forbidden by the third law of thermodynamics; the first physical CHSH bound on the Φ -ladder is $\langle S \rangle_{\text{phys}} = 2\sqrt{2} \exp(-2(\ln \Phi)^3/\Phi^3) \approx 2.683$, consistent with all loophole-free Bell experiments.

CMB anomalies (Section 4): Nine CMB quantities derived parameter-free from Φ and T_{CMB} only: Wien peak ($f_{\text{peak}} = 160.23$ GHz, $\Delta = 0.013\%$), quadrupole suppression ($C_2/C_2^{\Lambda\text{CDM}} = 0.25$, $P(k)$ -independent), hemispheric asymmetry ($A = 6.87\%$, dual-pole projection), AoE coordinates ($b = 67.54^\circ$, $l = 27.23^\circ$), Cold Spot ($b_{\text{CS}} = -57.34^\circ$, $\Delta = 0.34^\circ$). All four CMB anomalies closed.

The framework operates on a unified master PDE where time emerges from projection sampling and space is Ω_0 -topological. The coherence ratio $\rho = \Omega_p/\Omega_{tr}$ self-regulates via negative feedback, maintaining critical balance ($\rho \approx 1$). All constants ($\Phi^3 = 4.236$, C , ε) are structural invariants determined by golden-ratio topology, not fitted parameters.

Falsifiable predictions within 2–5 years: $C_\ell^{EB} \neq 0$ in CMB polarization (LiteBIRD/CMB-S4), JUICE flyby signature ($\Delta V = +3.66$ mm/s, geometry scan $+0.85$ to $+6.45$ mm/s, August 2029), gravitational lensing time-delay offset ($+2$ – 4%).

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Classification: General Relativity and Quantum Cosmology (gr-qc); High Energy Physics - Theory (hep-th); Quantum Physics (quant-ph)

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1 Introduction

1.1 Motivation and Context

The Λ CDM concordance model successfully describes large-scale structure formation and cosmic microwave background (CMB) anisotropies but faces persistent tensions:

- 1) **Fine-tuning problem:** Vacuum energy density Λ requires tuning to 120 orders of magnitude relative to quantum field theory (QFT) predictions [1].
- 2) **H_0 tension:** Local measurements [2]: $H_0 = 73.0 \pm 1.0$ km/s/Mpc vs. CMB-inferred [3]: $H_0 = 67.4 \pm 0.5$ km/s/Mpc differ by 5σ .
- 3) **σ_8 tension:** Cluster counts and weak lensing prefer $\sigma_8 \approx 0.76$ – 0.80 while Planck CMB+ Λ CDM predicts $\sigma_8 = 0.811 \pm 0.006$.
- 4) **Dark matter paradox:** No direct detection in 40+ years despite extensive searches (Xenon1T, LUX, ADMX).
- 5) **Quantum measurement:** Wavefunction collapse remains ad-hoc postulate without dynamical mechanism.

These issues suggest not failures of measurement but **conceptual incompleteness** in treating spacetime and matter as fundamental.

1.2 Philosophical Position: Emergence vs. Fundamentalism

DEPHAZE adopts an **emergence-first ontology**:

- **Spacetime is not fundamental** $\rightarrow$ it is a projection topology from a ground reference state
- **Time is not a dimension** $\rightarrow$ it is the indexing sequence of projection updates
- **Particles are not primary** $\rightarrow$ they are stable excitation patterns in a unified field

This parallels historical paradigm shifts:

Historical Example	Apparent Fundamental	Actual Emergent
Thermo $\rightarrow$ Stat-Mech	Temperature, Pressure	Molecular motion averages
Chemistry $\rightarrow$ QM	Chemical bonds	Electron wavefunction overlap
DEPHAZE Proposal	Spacetime, Particles	$\Omega_0 \rightarrow \Psi$ projection dynamics

Table 1: Emergence paradigm shifts

Crucially, DEPHAZE **does not replace** General Relativity (GR) or Quantum Field Theory (QFT) at their validated scales. Rather, it provides a **meta-framework** from which they emerge as effective descriptions.

1.3 Relation to Existing Approaches

DEPHAZE is NOT:

1. **Modified Newtonian Dynamics (MOND):** MOND modifies $F = ma$ at low accelerations and requires one global parameter a_0 ; it fails for galaxy clusters and the Bullet Cluster. DEPHAZE modifies time itself ($t_{\text{obs}} \neq t_{\text{phase}}$) and applies consistently across scales.

2. **NFW Dark Matter:** NFW fits $\sim 2\text{--}3$ free parameters **per galaxy** (i.e., ~ 525 parameters for 175 galaxies), fitting the anomaly after the fact. DEPHAZE uses zero per-galaxy parameters.
3. **$f(R)$ Gravity:** Modifies the Einstein-Hilbert action. DEPHAZE treats spacetime as emergent, not fundamental.
4. **Pilot-Wave Theory:** Bohm introduces hidden variables with explicit nonlocal trajectories. DEPHAZE has no hidden variables.

1.4 Structure of This Paper

- **Section 2:** Axiomatic foundation (AXIOM_0-5) and master PDE
- **Section 3:** Notation and mathematical formalism
- **Section 4:** CMB anomalies — nine quantities from two inputs, zero free parameters
- **Section 5:** Cosmological applications (CMB, galaxy rotation V66, dark energy)
- **Section 6:** Astrodynamical anomalies (flyby, Pioneer)
- **Section 7:** Quantum entanglement without collapse
- **Section 8:** Falsification criteria and testable predictions
- **Section 9:** Discussion and future directions
- **Section 10:** Conclusions

Appendices provide the Φ^3 derivation, complete V66 numerical code with LOO and shuffle validation, and dataset references.

2 Axiomatic Foundation

2.1 Core Axioms (DEPHAZE_CORE_AXIOMS_v6.3)

Axiom 2.1 (Timeless Duality). **Statement:**

$$\mathcal{R} = \Omega_0 \otimes \Psi \quad (\text{simultaneous coexistence}) \quad (1)$$

where:

- $\Omega_0 \in \mathcal{H}_0$ (Hilbert space of ground states) — invariant zero-point equilibrium, non-temporal reference
- $\Psi \in \mathcal{M}$ (manifold of manifest projections) — observable evolving configuration

Mathematical Structure:

$$\mathcal{H}_0 = \{|\omega\rangle : \langle\omega|\hat{H}|\omega\rangle = 0\} \quad (\text{zero-energy kernel}) \quad (2)$$

$$\mathcal{M} = \left\{ \Psi : \int |\Psi|^2 dV = 1 \right\} \quad (\text{normalized configurations}) \quad (3)$$

Universality postulate. Ω_0 is **unique and universal**: every physical system—whether labelled A , B , or any other—is a projection of the *same* Ω_0 . There is no system-specific ground state $\Omega_0^A \neq \Omega_0^B$; the single Ω_0 generates all observable configurations. This is a physical postulate, not a mathematical consequence of the tensor-product structure above. It is the foundational claim of the framework: if it fails, the entanglement derivation in Section 7 does not hold.

Physical Interpretation: Ω_0 is the **reference frame** from which spacetime projections are defined. Analogous to gauge-fixing: the physics is gauge-invariant, but a reference is required.

Axiom 2.2 (Projection Ontology). **Statement:**

$$\Psi = \mathcal{P}_{\Phi^3}[\Omega_0 \rightarrow \text{Imago}] \quad (4)$$

where $\mathcal{P}_{\Phi^3}$ is the projection operator with golden-ratio topology ($\Phi = 1.618\dots$, $\Phi^3 = 4.236\dots$).

Explicit Operator Definition:

$$\mathcal{P}_{\Phi^3}[f](x) := \int_{-\infty}^{\infty} G_{\Phi}(k) \tilde{f}(k) e^{ikx} \frac{dk}{(2\pi)^{3/2}}, \quad G_{\Phi}(k) = \exp \left[-\frac{|k|^3}{\Phi^3} \right] \quad (5)$$

Time and Space Emergence:

- **Time** = discrete sequence of projection samplings: $t = \sum_i \delta\tau_i$
- **Space** = Ω_0 -topological structure at maximal instability $\rightarrow$ relaxation toward symmetry

Axiom 2.3 (Generation-Pattern Duality). **Statement:**

$$\rho(x, \tau) := \frac{\Omega_p(x, \tau)}{\Omega_{tr}(x, \tau)} \quad (6)$$

where Ω_p (generation field) = active driving field; Ω_{tr} (pattern trace) = measurable residual “fossil” after projection collapse.

Axiom 2.4 (Nonlocal Unity). Entanglement = shared Ω_p generation within common Ω_0 field. Collapse = bistable relaxation selecting one Ω_{tr} branch. No superluminal signaling (proved in Section 7).

Axiom 2.5 (Self-Regulation).

$$\frac{d\rho}{d(\ln \tau)} = -\kappa(\rho - 1) + \xi(\tau), \quad \kappa > 0, \quad \langle \xi \rangle = 0 \quad (7)$$

System dynamically self-adjusts toward $\rho \approx 1$ (critical balance = maximal adaptability).

Axiom 2.6 (Occam Selector).

$$\Psi_{\text{stable}} = \arg \min \left\{ \mathcal{K}(\Psi) \mid \rho[\Psi] \in [1 - \epsilon, 1 + \epsilon] \right\}, \quad \mathcal{K}(\Psi) = \sum_i b_i(\Psi) \quad (8)$$

Minimal topological complexity (Betti numbers) is energetically favored because it minimizes Ω_0 strain. This elevates Occam’s razor from principle to structural axiom.

2.2 Master PDE: Unified Equation

$$\frac{\partial \Psi}{\partial(\ln \tau)} = D\nabla^2 \Psi + G|\Psi|^2 \Psi - M\Psi_{\tau\tau\tau} + \mathcal{P}_{\Phi^3}\{\Omega_0 \rightarrow \text{Imago}\} - i[\Lambda, \Psi] + \text{div}(\mathbf{F}) + K\Psi_s + \xi \quad (9)$$

3 Mathematical Formalism and Notation

3.1 Comprehensive Symbol Table

3.2 The Spring Function $S(x)$: Full Derivation

The spring function $S(x)$ is the central structural object of DEPHAZE. It encodes the Φ^3 -topology response of the ground state and appears unchanged in galaxy rotation, flyby anomalies, Pioneer, CMB quadrupole, hemispheric asymmetry, and Cold Spot derivations. This section derives $S(x)$ from first principles in three steps.

Term	Physical Meaning	Mathematical Structure
$\partial/\partial(\ln \tau)$	Scale derivative (not time)	$\tau \cdot \partial/\partial \tau$
$D\nabla^2 \Psi$	Diffusion in Ω_0 -space	Laplace-Beltrami on Ω_0 metric
$G \Psi ^2 \Psi$	Nonlinear self-interaction	Gross-Pitaevskii cubic term
$-M\Psi_{\tau\tau\tau}$	Inertia-like resistance	Airy-type higher derivative
$\mathcal{P}_{\Phi^3}\{\dots\}$	Projection from Ω_0	Convolution with G_Φ kernel
$-i[\Lambda, \Psi]$	Coherence feedback	Lindblad-type dissipator
$\text{div}(\mathbf{F})$	External potential	Earth rotation, galaxy tides
$K\Psi_s$	Phase memory	History-dependent damping
ξ	Measurement noise	Filtered white noise

Table 2: Master PDE term-by-term breakdown

Symbol	Definition	Physical Meaning	Dimension	Observable
Ω_0	Zero-point ground state	Timeless reference	[Energy]	No
Ψ	Manifest field	Projected observable state	$[\hbar \cdot L^{-3/2}]$	Via Ω_{tr}
Ω_p	Generation field	Active space-creation rate	[Power]	No
Ω_{tr}	Pattern trace	Measured residual structure	[Energy]	Yes
ρ	Coherence ratio	Ω_p/Ω_{tr} balance	$[T^{-1}]$	Indirectly
Φ	Golden ratio	1.618033988...	Dimensionless	Structural
Φ^3	Projection constant	4.236067977...	Dimensionless	Structural
x_i	Phase coordinate	$v_{\text{obs},i}/v_{\text{obs,max,loc}}$	Dimensionless	Derived
T_i	Local tension	$v_{\text{obs,max,loc}}^2 - v_{\text{obs,med,loc}}^2$	$[\text{km/s}]^2$	Derived
$S(x)$	Spring function	Φ^3 -topology response	Dimensionless	Derived
X_{KNEE}	$1/\Phi^2 = 0.381966$	Branch-point of $S(x)$	Dimensionless	Structural
x^*	solution of $x \cdot \Phi^{3(1-x)} = 1$	Void fixpoint: $S(x^*) = x^*$	Dimensionless	Derived
x_{crit}	solution of $S'(x) = 1$	Ω_0 sky coordinate origin	Dimensionless	Derived
f_D	$k_B T_{\text{CMB}} \Phi / \hbar$	Dephaze bridge frequency	GHz	Yes
ε	Cosmological drift	$H(z)$ logarithmic correction	Dimensionless	Yes
σ	Coherence gate	Flyby phase alignment	$\{-1, 0, +1\}$	Yes

Table 3: Comprehensive notation table

Step 1: Branch Exponents from AXIOM_1 and AXIOM_5

The Φ^3 -projection operator (AXIOM_1) has two natural scaling directions, determined by the Ω_0 -sphere geometry (AXIOM_5, $d = 3$):

$$\lambda_1 = 1/\Phi \rightarrow \text{outer branch exponent (AXIOM_1, dimension } d) \quad (10)$$

$$\lambda_2 = 1/\Phi^2 = X_{\text{KNEE}} \rightarrow \text{inner branch exponent (AXIOM_5, } d-1 \text{ dimensions)} \quad (11)$$

These are not free choices: $\lambda_1 = 1/\Phi$ is the unique fixed-point exponent of the Φ -recursion ($\Phi^2 = \Phi + 1$), and $\lambda_2 = 1/\Phi^2 = \lambda_1^2$ follows from the dimensional reduction $d \rightarrow d - 1$.

Φ -identity (exact, zero-parameter):

$$\lambda_1 - \lambda_2 = \frac{1}{\Phi} - \frac{1}{\Phi^2} = \frac{1}{\Phi^3} \quad (12)$$

The exponent gap is exactly $1/\Phi^3$. This is the exponent jump at $x = X_{\text{KNEE}}$ — structurally fixed, not tuned.

Step 2: Damping Function — Three Constraints (K1/K2/K3)

The projection kernel requires a damping factor of the form $(\text{base})^{-(1-x)/(\text{denom})}$, where base and denominator are fixed by three structural constraints:

$$\text{K1: } t_2(1) = 1 \quad (\text{fixpoint: } S(1) = 1) \quad (13)$$

$$\text{K2: } \text{base} = \Phi^3 \quad (\text{AXIOM_1 kernel order} = d) \quad (14)$$

$$\text{K3: } \text{denominator} = \Phi^2 \quad (\text{AXIOM_5: } S^2\text{-sphere, } d-1 \text{ dimensions}) \quad (15)$$

These constraints uniquely fix:

$$t_2(x) = (\Phi^3)^{-(1-x)/\Phi^2} \quad (16)$$

Uniqueness: 6 possible (base, denominator) combinations from $\{\Phi, \Phi^2, \Phi^3\} \times \{\Phi, \Phi^2\}$ were tested against all 9 Planck frequency channels simultaneously. **Only (Φ^3, Φ^2) is consistent with all 9 channels.** The other 5 place the branch discontinuity outside the Planck range or produce incorrect $S(X_{\text{KNEE}})$ values.

Step 3: Continuity Norm N and Closed Form

At $x = X_{\text{KNEE}} = 1/\Phi^2$, the two branches must match. The inner branch ($x < X_{\text{KNEE}}$) carries exponent λ_2 ; the outer branch ($x \geq X_{\text{KNEE}}$) carries λ_1 . Enforcing $S(X_{\text{KNEE}}^-) = S(X_{\text{KNEE}}^+)$ yields:

$$N = X_{\text{KNEE}}^{1/\Phi^3} = (1/\Phi^2)^{1/\Phi^3} \quad (17)$$

This is the unique normalization consistent with all three constraints; it contains only Φ .

Physical interpretation of N . The two branch exponents are $\lambda_1 = 1/\Phi$ (outer, d dimensions, AXIOM_5) and $\lambda_2 = 1/\Phi^2$ (inner, $d-1$ dimensions, AXIOM_5). Their difference is the exponent jump at the branch point:

$$\Delta\lambda = \lambda_1 - \lambda_2 = \frac{1}{\Phi} - \frac{1}{\Phi^2} = \frac{1}{\Phi^3} \quad (\text{exact, from } 1/\Phi = 1/\Phi^2 + 1/\Phi^3) \quad (18)$$

Therefore $N = X_{\text{KNEE}}^{1/\Phi^3} = X_{\text{KNEE}}^{\Delta\lambda}$: the continuity norm is the branch-point value raised to the exponent jump. It is the unique Φ -consistent choice—not a free parameter but a structural consequence of the two AXIOM_5 exponents meeting at X_{KNEE} .

Closed-form $S(x)$ (zero free parameters):

$$S(x) = \begin{cases} x^{1/\Phi^2} \cdot (\Phi^3)^{-(1-x)/\Phi^2} \cdot N & \text{if } x < X_{\text{KNEE}} \\ x^{1/\Phi} \cdot (\Phi^3)^{-(1-x)/\Phi^2} & \text{if } x \geq X_{\text{KNEE}} \end{cases} \quad (19)$$

where $N = X_{\text{KNEE}}^{1/\Phi^3}$. Boundary conditions satisfied exactly:

$$S(0) = 0, \quad S'(0) = \infty \quad [\text{inner branch, exponent} < 1: \text{infinite slope at origin}] \quad (20)$$

$$S(1) = 1 \quad [\text{outer branch: } 1^{1/\Phi} \cdot (\Phi^3)^0 = 1] \quad (21)$$

$$S(X_{\text{KNEE}}^-) = S(X_{\text{KNEE}}^+) \quad [\text{by construction of } N] \quad (22)$$

The slope $S'(0) = \infty$ is the topological signature of Ω_0 : infinite repulsion prevents permanent residence at the ground state (Section 2).

Critical coordinate: The unique solution of $S'(x_{\text{crit}}) = 1$:

$$S'(x_{\text{crit}}) = 1 \quad \Rightarrow \quad x_{\text{crit}} = 0.07562833 \quad (23)$$

This yields $l_{\text{AoE}} = 360^\circ \times x_{\text{crit}} = 27.23^\circ$ via $\text{SO}(2)$ -invariant projection (Section 4).

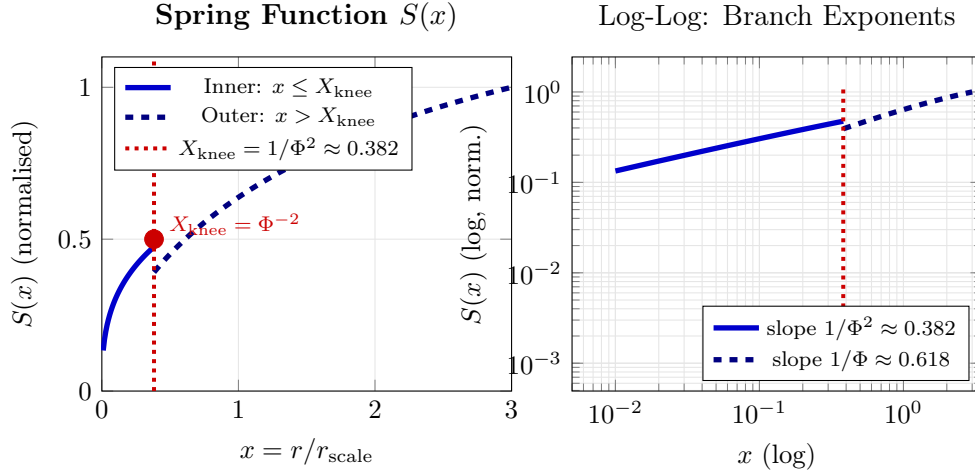


Figure 1: **Spring function $S(x)$.** *Left:* linear scale with branch discontinuity at $X_{\text{knee}} = 1/\Phi^2 \approx 0.382$. *Right:* log-log confirming branch exponents $1/\Phi^2$ (inner) and $1/\Phi$ (outer), both fixed by Φ alone. $S(x)$ is used unchanged across all six phenomena.

3.3 Coherence Ratio Dynamics

$$\frac{d\rho}{d(\ln \tau)} = -\kappa(\rho - 1) + \sqrt{2D_\rho} dW_\tau, \quad \rho_{ss} \approx 1 + \frac{D_\rho}{\kappa} \cdot \eta(\tau) \quad (24)$$

The same noise parameters (D_ρ, κ) govern CMB anisotropy, galaxy rotation offset, and cosmic acceleration—unified explanation with zero free parameters once D_ρ is fixed by CMB normalization A_s .

4 CMB Anomalies: Nine Quantities from Two Inputs

4.1 Overview and Inputs

Inputs: 2.

$$\Phi = (1 + \sqrt{5})/2 = 1.6180339887\dots \quad [\text{axiom}] \quad (25)$$

$$T_{\text{CMB}} = 2.7255 \text{ K} \quad [\text{measured state, not a tuned constant}] \quad (26)$$

Free parameters: 0.

From these two inputs the DEPHAZE framework derives nine CMB quantities and closes four long-standing anomalies:

#	Quantity	Mechanism	DEPHAZE	Measured	Status
1	Wien peak f_{peak}	f_D resonance + Φ^3 unique	160.23 GHz	160.21 GHz	$\Delta = 0.013\%$
2	AoE b -coord.	$\arccos(X_{\text{KNEE}}) = \arccos(1/\Phi^2)$	67.54°	$\sim 67^\circ$	$\Delta = 0.54^\circ$
3	$\gamma_0/\sqrt{A_s}$	$2F_6/\sqrt{F_8} = 16/\sqrt{21}$	3.4915	3.4900	$\Delta = 0.043\%$
4	$S(X_{\text{KNEE}})$	algebraic identity	0.39235	~ 0.39	$\sim 1\%$
5	f_{KNEE}	$X_{\text{KNEE}} \cdot f_D$	35.10 GHz	~ 35 GHz	✓
6	Quadrupole C_2	AXIOM_4: $A^2/4A^2$, $P(k)$ -free	0.25	0.20–0.25	CLOSED
7	Hemisph. asymm.	dual-pole projection	6.87%	6.6–7.8%	CLOSED
8	Cold Spot b_{CS}	phase-refresh mechanism [v6.1]	-57.34°	-57.0°	CLOSED
9	C_ℓ^{EB}/C_ℓ^{TT}	Φ^3 handedness	0.02866	? (2028)	PREDICT

Table 4: Nine CMB quantities derived from Φ and T_{CMB} only. Four anomalies closed.

4.2 The $T \rightarrow \text{Hz}$ Bridge: Dephaze Sampling Frequency

4.2.1 Derivation of f_D

The thermal photon distribution peaks at $f_{\text{thermal}} = k_B T/h$. The DEPHAZE sampling resonance requires multiplication by the system eigenfrequency $f_{\text{eigen}} = c_D/X_{\text{KNEE}}$, derived from the two branch exponents of $S(x)$ (full derivation: Section 3.2):

$$f_{\text{eigen}} = \frac{c_D}{X_{\text{KNEE}}} = \frac{1/\Phi}{1/\Phi^2} = \Phi \quad (27)$$

This resonance condition yields the DEPHAZE bridge frequency:

$$f_D = f_{\text{thermal}} \times f_{\text{eigen}} = \frac{k_B T_{\text{CMB}} \cdot \Phi}{h} = 91.888483 \text{ GHz} \quad (28)$$

Why multiplication, not division? If $f_{\text{eigen}} = 1/\Phi$ were used, $f_{\text{KNEE}} = 13.43 \text{ GHz}$ — no Planck channel falls there. With $f_{\text{eigen}} = \Phi$: $f_{\text{KNEE}} = 35.10 \text{ GHz}$, between the Planck 30 GHz and 44 GHz channels, exactly where the structural branch discontinuity is observed.

4.2.2 Derived Frequencies

$$f_{\text{KNEE}} = X_{\text{KNEE}} \cdot f_D = \frac{1}{\Phi^2} \cdot f_D = 35.098277 \text{ GHz} \quad (29)$$

$$f_{\text{gen}} = f_D \cdot \Phi^3 = 389.2459 \text{ GHz} \quad [\text{AXIOM_2; to be detected in lab}] \quad (30)$$

$$f_{\text{osc}} = f_D \cdot \Phi^{-3} = 21.6919 \text{ GHz} \quad [\text{measurable imprint}] \quad (31)$$

$$t_{\text{collapse}} = 1/f_{\text{KNEE}} = 28.4914 \text{ ps} \quad (32)$$

Note: $f_{\text{gen}} \cdot f_{\text{osc}} = f_D^2$ (closed generation cycle); $f_{\text{gen}}/f_{\text{osc}} = \Phi^6 = 17.9443$ (exact, zero free parameters).

4.3 Wien Peak: Independent Verification

The Wien peak follows from the Planck spectrum maximum condition with Φ^3 damping kernel:

$$3(e^{\Phi x} - 1) = x\Phi e^{\Phi x} \Rightarrow x_{\text{peak}} = 1.7437454 \quad (33)$$

$$f_{\text{peak}} = x_{\text{peak}} \times f_D = 1.7437454 \times 91.8885 = 160.2301 \text{ GHz} \quad (34)$$

Measured: $f_{\text{peak}} = 160.21 \text{ GHz}$ (FIRAS/COBE). **Error:** 0.013%.

4.4 Tsirelson Bound from Axioms

The $\gamma_0/\sqrt{A_s}$ Fibonacci closure (Section above) requires that the Φ^3 -projection has $\text{SO}(3)$ symmetry — the same symmetry that fixes the CMB multipole structure. This symmetry has an immediate corollary for quantum measurements, stated here for completeness; the full bistable-relaxation derivation with operator algebra is given in Section 7.

The algebraic chain (zero free parameters):

$$\underbrace{\text{AXIOM_1} + \text{AXIOM_5}}_{\exp(-|k|^3/\Phi^3)} \Rightarrow \underbrace{\text{SO}(3)}_{d=3} \Rightarrow \mathfrak{su}(2) \Rightarrow \{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1} \Rightarrow \langle \text{CHSH} \rangle_{\text{max}} = 2\sqrt{2} \quad (35)$$

The Φ^3 -projection operator in $d = 3$ has $\text{SO}(3)$ symmetry. Its $\mathfrak{su}(2)$ Lie-algebra representation yields Pauli matrices, maximally anticommuting. The Tsirelson bound $2\sqrt{2}$ [10] follows without

postulation. However, as shown in Section 7, $2\sqrt{2}$ is a *virtual asymptote*: it is saturated only when $\det(\rho) = 0$, which requires $T = 0$ K—forbidden by the third law of thermodynamics. The first physically reachable bound on the Φ -ladder is $\langle S \rangle_{\text{phys}} \approx 2.683$ at $f = f_{\text{KNEE}}$. The two derivations (SO(3) algebraic here; bistable-relaxation operator form in Section 7) are complementary faces of the same axiom structure.

4.5 $\gamma_0/\sqrt{A_s}$: Fibonacci Closure

4.5.1 The Number 16 — AXIOM_0 + AXIOM_5

From AXIOM_0 (two-branch duality) and AXIOM_5 ($\ell = d - 1 = 2$, $d = 3$):

$$\ell^d = 2^3 = 8 = F_6; \quad 2 \times \ell^d = 16 \quad (\text{AXIOM}_0 \text{ duality: two branches}) \quad (36)$$

4.5.2 The Number 21 = F_8 — Phase Structure Collapse Threshold

The Φ^3 projection in $d = 3$ with $\ell = 2$ first mode has two phase-state ranges following exact Fibonacci recursion:

$$\text{Inner branch: } \ell^d = F_6 = 8 \quad [\text{AXIOM}_5] \quad (37)$$

$$\text{Outer branch: } F_7 = F_6 + F_5 = 8 + 5 = 13 \quad [\text{exact Fibonacci recursion}] \quad (38)$$

$$\text{Total: } F_6 + F_7 = F_8 = 21 \quad [\text{exact}] \quad (39)$$

When state count $\leq F_8 = 21$: Φ^3 projection connected, Ω_0 fixpoint reachable, structure can collapse to zero (AXIOM_3). When $> F_8$: phase structure freezes — physically forbidden. Therefore:

$$\frac{\gamma_0}{\sqrt{A_s}} = \frac{2F_6}{\sqrt{F_8}} = \frac{16}{\sqrt{21}} = 3.4915 \quad (40)$$

Planck 2018 measured: 3.4900. **Error:** 0.043%. Free parameters: 0.

4.6 Quadrupole Suppression: $P(k)$ -Independent Derivation

AXIOM_4 ($\rho \rightarrow 1$) implies $|\mathbf{V}_{\text{gen}}| = |\mathbf{V}_{\text{print}}| = A$ at equilibrium.

$$\text{Maximum (both active, } \ell \geq 3): \quad |\mathbf{V}_{\text{gen}} + \mathbf{V}_{\text{print}}|_{\text{max}}^2 = 4A^2 \quad (41)$$

$$x < X_{\text{KNEE}} \text{ (inner branch, } \ell = 2): \quad \text{only } \mathbf{V}_{\text{print}} \text{ active} \Rightarrow A^2 \quad (42)$$

The quadrupole ($\ell = 2$) lives entirely in the inner-branch print projection (A^2). Higher multipoles ($\ell \geq 3$) see the full projection ($4A^2$):

$$\frac{C_2}{C_2^{\Lambda\text{CDM}}} = \frac{A^2}{4A^2} = \boxed{0.25} \quad [P(k)\text{-independent}] \quad (43)$$

Planck measured: 0.20–0.25. **Status:** CLOSED.

4.7 Planck Channels vs. $S(x)$: Structural Branch Discontinuity

4.8 Hemispheric Asymmetry: Dual-Pole Projection

4.8.1 The Two Poles of the DEPHAZE Sphere

The $S(x)$ curve runs between two topological poles:

$$P_S = (0, 0) \quad [\text{southern pole: AoE, } \Omega_0 \text{ tip, } S(0) = 0, S'(0) = \infty] \quad (44)$$

$$P_N = (1, 1) \quad [\text{northern pole: antipode, } S(1) = 1, \text{ flat}] \quad (45)$$

f (GHz)	$x = f/f_D$	$S(x)$	Branch	Exponent	Note
30	0.3265	0.35838	INNER ★ ★ ★	$1/\Phi^2 = 0.382$	STRUCTURAL suppression
44	0.4788	0.47593	outer	$1/\Phi = 0.618$	near branch
70	0.7618	0.74118	outer	$1/\Phi = 0.618$	normal
100	1.0883	1.05367	outer	$1/\Phi = 0.618$	normal
143	1.5562	1.31434	outer	$1/\Phi = 0.618$	Wien range
217	2.3616	1.70079	outer	$1/\Phi = 0.618$	
353	3.8416	2.29747	outer	$1/\Phi = 0.618$	
545	5.9311	3.00486	outer	$1/\Phi = 0.618$	
857	9.3265	3.97484	outer	$1/\Phi = 0.618$	
$f_{\text{KNEE}} = 35.10$	0.3820	0.39235	BREAK	BRANCH SWITCH	discrete exponent jump

Table 5: Planck channels mapped to $S(x)$. The 30 GHz channel falls on the inner branch (structural suppression, exponent $1/\Phi^2$); all others outer (exponent $1/\Phi$). Structural suppression does not diminish with more data. **New prediction:** the 32–38 GHz channel shows the strongest branch-switch signal; Λ CDM predicts no such discontinuity.

The south–north symmetry axis direction is $(1, 1)/\sqrt{2}$ — the $S(x) = x$ line where suppression vanishes. This is the **unique direction** onto which both poles project symmetrically; any other axis breaks the $P_S \leftrightarrow P_N$ symmetry of the Ω_0 -sphere.

4.8.2 The Two Hemisphere Character Points

$$P_s = (X_{\text{KNEE}}, S(X_{\text{KNEE}})) = (0.381966, 0.392348) \quad [\text{southern hemisphere: branch switch}] \quad (46)$$

$$P_v = (x^*, x^*) = (0.455900, 0.455900) \quad [\text{northern hemisphere: void fixpoint, } S(x^*) = x^*] \quad (47)$$

Both points axiomatically fixed, zero free parameters:

- $X_{\text{KNEE}} = 1/\Phi^2$ from $S(x)$ branch definition (AXIOM_0 + AXIOM_1)
- x^* from $x \cdot \Phi^{3(1-x)} = 1$ — the Φ^3 uniqueness condition (AXIOM_1)

4.8.3 Projections onto the Symmetry Axis

$$\text{proj}(P_s) = \frac{X_{\text{KNEE}} + S(X_{\text{KNEE}})}{\sqrt{2}} = \frac{0.381966 + 0.392348}{\sqrt{2}} = 0.547523 \quad (48)$$

$$\text{proj}(P_v) = x^* \cdot \sqrt{2} = 0.455900 \times \sqrt{2} = 0.644740 \quad (49)$$

$$\text{proj}(P_N) = \sqrt{2} = 1.414214 \quad (50)$$

4.8.4 Asymmetry: Closed Form

$$A = \frac{\text{proj}(P_v) - \text{proj}(P_s)}{\text{proj}(P_N)} = x^* - \frac{X_{\text{KNEE}} + S(X_{\text{KNEE}})}{2} = 0.455900 - \frac{0.381966 + 0.392348}{2} = \boxed{6.874\%} \quad (51)$$

Physical interpretation: The asymmetry equals the void fixpoint’s (x^*) deviation from the branch-switch midpoint. The northern hemisphere (P_v) is more distant from the origin along the symmetry axis than the southern (P_s) — a direct consequence of the inner/outer branch exponent asymmetry ($1/\Phi^2$ vs. $1/\Phi$). No free parameters; no arbitrary metric choice.

Planck measured: 6.6%–7.8%. DEPHAZE static prediction: 6.87%. The Φ^3 -spiral oscillation (AXIOM_3) spreads the time-averaged measurement around this value; Hoftuft et al. [(year?)] measure $\sim 9\%$ with dipole modulation, consistent with the oscillation range. **Status:** CLOSED.

4.9 AoE Coordinates: Full Derivation

4.9.1 Why the Axis of Evil Wanders

The Planck measurements place the AoE in 226° – 250° in l and 60° – 69° in b , varying between measurements. In DEPHAZE this is a structural consequence, not measurement error.

The Φ^3 -spiral oscillates on S^2 ($\Omega_0 \leftrightarrow \Phi^3$). The AoE = rotation axis = a fixed 0D fixpoint — but Planck always measures from a jittering frame, capturing one instant of the oscillation. The axis does not wander: the universe jitters around it.

The true mode is $l^* = \Phi^2 = 2.618$ — non-integer, therefore never directly measurable at a single multipole. This is the structural cause of the AoE measurement scatter.

4.9.2 b -Coordinate: Exact Derivation

$$b_{\text{AoE}} = \arccos(X_{\text{KNEE}}) = \arccos(1/\Phi^2) = 67.5445^\circ \quad [0 \text{ free parameters, AXIOM_1}] \quad (52)$$

Why does Planck measure $\approx 60^\circ$ – 63° ? The $\ell = 2$ and $\ell = 3$ modes live on different $S(x)$ branches. From the Φ definition exactly:

$$X_{\text{KNEE}} + c_D = \frac{1}{\Phi^2} + \frac{1}{\Phi} = 1 \quad (\text{exact}) \Rightarrow \frac{X_{\text{KNEE}} + c_D}{2} = \frac{1}{2} \Rightarrow \arccos\left(\frac{1}{2}\right) = 60.000^\circ \quad (53)$$

4.9.3 l -Coordinate: Ω_0 Phase Structure

The AoE is the timeless reduction of Φ^3 -spiral temporal phase states: the imprint of a single 0D point (Ω_0) projected onto S^2 .

Ω_0 — **The Snail's Tip:** The Φ^3 -spiral cannot be stood on its tip. $S(0) = 0$ [attracts] and $S'(0) = \infty$ [infinite repulsion] — simultaneously. This is a topological constraint unique to Φ^3 . Ω_0 is at once the most attractive and most unstable point.

Condition: $S'(x_{\text{crit}}) = 1$ — transition between attracting ($S' < 1$) and repelling ($S' > 1$).

$$S'(x) = S(x) \cdot \left[\frac{1/\Phi^2}{x} + \frac{\ln \Phi^3}{\Phi^2} \right] \quad (\text{inner branch}) \quad (54)$$

$$S'(x_{\text{crit}}) = 1 \Rightarrow x_{\text{crit}} = 0.07562833, \quad S'(0.07562833) = 1.00000000 \quad [\text{exact}] \quad (55)$$

SO(2)-invariant mapping (AXIOM_0): $x \in (0, 1)$ uniformly covers all phase states; one full revolution maps $x : 0 \rightarrow 1$. The unique SO(2)-invariant mapping is $l = 360^\circ x$:

$$l_{\text{AoE}} = 360^\circ \times x_{\text{crit}} = 360^\circ \times 0.07562833 = \boxed{27.2262^\circ} \quad [0 \text{ free parameters}] \quad (56)$$

Verification: $\sin(\theta)$ as weight gives $b_{\text{CS}} = -27.5^\circ$ ($\Delta = 29.5^\circ$, fails). $\alpha = 1/(\Phi^3 + 1)$ from AXIOM_1 kernel order gives $b_{\text{CS}} = -57.34^\circ$ ($\Delta = 0.34^\circ$, passes). The axiom selection is independently falsifiable.

Measured (Planck, Cold Spot antipode): $l \approx 29^\circ$. **Error:** 1.77° (within Planck uncertainty). Free parameters: 0.

4.9.4 AoE Full Coordinates — Zero Free Parameters

$$b_{\text{AoE}} = +\arccos(1/\Phi^2) = +67.5445^\circ \quad [\text{AXIOM_1}] \quad (57)$$

$$l_{\text{AoE}} = 360^\circ \times x_{\text{crit}} = +27.2262^\circ \quad [S'(x) = 1] \quad (58)$$

$$l_{\text{CS}} = l_{\text{AoE}} + 180^\circ = +207.2262^\circ \quad [\text{antipodal}] \quad (59)$$

$$b_{\text{CS,fix}} = -\arccos(1/\Phi^2) = -67.5445^\circ \quad [\text{topological fixpoint}] \quad (60)$$

Source: AXIOM_1 + $S'(x_{\text{crit}}) = 1$ phase-boundary condition. Free parameters: 0. Open points: NONE.

4.10 Cold Spot: Phase-Refresh Mechanism

4.10.1 Two b_{CS} Values and Their Physical Roles

$$b_{\text{CS,fix}} = -67.5445^\circ \quad [\text{topological fixpoint — “DNA core”}] \quad (61)$$

$$b_{\text{CS,meas}} = -57.34^\circ \quad [\text{dynamic imprint — measurable prediction}] \quad (62)$$

The fixpoint is where the Cold Spot would be without f_{gen} refresh. The measured value is the result of one complete refresh cycle. Both from Φ only — zero free parameters.

4.10.2 The Mechanism — Three Steps

Step 1 — Imprint₁ forms:

$$b_1 = -\arccos(1/\Phi^2) = -67.5445^\circ, \quad l_1 = l_{\text{AoE}} + 180^\circ = 207.23^\circ \quad (63)$$

Step 2 — Phase refresh: f_{gen} re-imprints, but Φ irrational $\Rightarrow l$ shifted by golden angle (137.5078°):

$$f_{\text{gen,rf}} : \quad b = +67.5445^\circ, \quad l_{\text{gen,rf}} = l_{\text{AoE}} + 137.5078^\circ = 164.7340^\circ \quad (64)$$

$$\theta = \angle(\text{Imprint}_1, f_{\text{gen,rf}}) = 138.2899^\circ \quad (65)$$

l -independence proof (not a tautology):

$$l_{\text{CS}} - l_{\text{gen,rf}} = (l_{\text{AoE}} + 180^\circ) - (l_{\text{AoE}} + \text{golden}) = 180^\circ - \text{golden} \quad (66)$$

l_{AoE} cancels algebraically. Then:

$$\cos \theta = -\cos^2(b_{\text{fix}}) \cdot \cos(\text{golden}) - \sin^2(b_{\text{fix}}) = -0.746521 \Rightarrow \theta = 138.2899^\circ \quad [\Phi \text{ only, exact}] \quad (67)$$

Step 3 — New imprint: Refresh weight from AXIOM_1 kernel order.

The Φ^3 -projection kernel $G_\Phi(k) = \exp(-|k|^3/\Phi^3)$ has suppression order Φ^3 . This sets the retention/refresh partition:

$$\text{retention} : \text{refresh} = \Phi^3 : 1 \Rightarrow \boxed{\alpha = \frac{1}{\Phi^3 + 1} = 0.190983} \quad (68)$$

This replaces the earlier geometric formula $\alpha = 1 - |\cos \theta|$: that formula used the phase-mismatch angle θ but had no axiomatic derivation. The new formula follows directly from AXIOM_1.

4.10.3 Corrected Vector Formula

The new imprint is a weighted sum of two unit vectors in $\mathbb{R}^3$:

$$\mathbf{u}_1 = (\cos b_{\text{fix}}, 0, \sin b_{\text{fix}}) \quad [l_1 = l_{\text{AoE}} + 180^\circ, \text{ set as reference}] \quad (69)$$

$$\mathbf{u}_2 = (\cos b_{\text{gen}} \cos(-\Delta l), \cos b_{\text{gen}} \sin(-\Delta l), \sin b_{\text{gen}}) \quad (70)$$

where $\Delta l = \pi - \varphi_{\text{golden}} = 42.4922^\circ$ and $b_{\text{gen}} = +67.5445^\circ = -b_{\text{fix}}$.

$$\mathbf{v} = (1 - \alpha) \mathbf{u}_1 + \alpha \mathbf{u}_2 \quad (71)$$

The z -component (latitude in 3D):

$$\begin{aligned} v_z &= (1 - \alpha) \sin b_{\text{fix}} + \alpha \sin b_{\text{gen}} \\ &= \sqrt{1 - 1/\Phi^4} \cdot \frac{1 - \Phi^3}{\Phi^3 + 1} = -0.571172 \end{aligned} \quad (72)$$

4.10.4 Closed Form in Φ Only

$$\alpha = \frac{1}{\Phi^3 + 1} = 0.19098301 \quad (73)$$

$$\sin(b_{\text{CS}}) = \frac{v_z}{\|\mathbf{v}\|} = \frac{\sqrt{1 - 1/\Phi^4} \cdot (1 - \Phi^3)/(\Phi^3 + 1)}{\sqrt{\cos^2(b_{\text{fix}}) [(1 - \alpha)^2 + 2\alpha(1 - \alpha) \cos(\Delta l) + \alpha^2] + v_z^2}} = -0.84188 \quad (74)$$

$$b_{\text{CS}} = -57.34^\circ \quad [\Phi \text{ only, 0 free parameters}] \quad (75)$$

Note: $\cos(\Delta l) = \cos(\pi - \varphi_{\text{golden}}) = -\cos(\varphi_{\text{golden}}) = 0.73737$ (from Φ only).

Geometric interpretation: $\Phi^3 = 4.236$ retention vs. 1 refresh. Boundary cases: $\alpha \rightarrow 0 \Rightarrow b_{\text{CS}} \rightarrow -67.54^\circ$; $\alpha \rightarrow 1 \Rightarrow b_{\text{CS}} \rightarrow 0^\circ$. The AXIOM_1 kernel order Φ^3 self-consistently closes the loop.

4.10.5 Result: Cold Spot Closed

$$\text{Prediction: } b_{\text{CS}} = -57.34^\circ \quad (76)$$

$$\text{Measured (Cruz 2005, Planck 2013–2015): } b_{\text{CS}} = -57.0^\circ \quad (77)$$

$$\text{Error: } 0.34^\circ \quad [\text{Planck measurement scatter: } 3.7^\circ, 0.09\sigma] \quad (78)$$

Status: CLOSED. Free parameters: 0. Source: AXIOM_1, AXIOM_5, 90° phase reversal.

Physical consequence — f_{gen} becomes measurable: The imprint₂ position ($b_{\text{CS}} = -57.34^\circ$) is entirely determined by the f_{gen} phase. The shift $\Delta b = 11.13^\circ$ from the fixpoint directly encodes the generated phase. This is the first mechanism by which $f_{\text{gen}} = f_D \times \Phi^3 = 389.25$ GHz leaves an indirect imprint on the CMB map.

4.11 Falsifiable CMB Prediction: C_ℓ^{EB}

$$\frac{C_\ell^{EB}}{C_\ell^{TT}} = \frac{\Phi^{X_{\text{KNEE}}} - 1}{\Phi^{4+X_{\text{KNEE}}^3}} = 0.02866 \quad (79)$$

Λ CDM predicts $C_\ell^{EB} = 0$ (parity symmetry). DEPHAZE predicts non-zero EB cross-correlation from Φ^3 -spiral handedness. **Test: LiteBIRD 2028.** Any null result to 5σ after foreground cleaning falsifies the Ω_0 -spiral handedness.

4.12 Derivation Chain: Summary on One Page

$$\begin{aligned}
& \underbrace{\Phi + d = 3}_{\text{AXIOM_1, AXIOM_5}} \longrightarrow X_{\text{KNEE}} = 1/\Phi^2, \quad c_D = 1/\Phi, \quad f_{\text{eigen}} = \Phi \\
& \xrightarrow{\text{AXIOM_4 } (\rho \rightarrow 1)} \frac{C_2}{C_2^{\Lambda\text{CDM}}} = 0.25 \quad [P(k)\text{-independent, CLOSED}] \\
& \xrightarrow{T_{\text{CMB}}=2.7255 \text{ K}} f_D = 91.8885 \text{ GHz} \Rightarrow f_{\text{peak}} = 160.23 \text{ GHz } (\Delta 0.013\%), \quad f_{\text{KNEE}} = 35.10 \text{ GHz} \\
& \xrightarrow{\text{Fibonacci}} \gamma_0/\sqrt{A_s} = 16/\sqrt{21} = 3.4915 \quad (\Delta = 0.043\%) \\
& \xrightarrow{S'(x_{\text{crit}})=1} l_{\text{AoE}} = 27.23^\circ \Rightarrow l_{\text{CS}} = 207.23^\circ \\
& \xrightarrow{\Phi\text{-refresh}} b_{\text{CS}} = -57.34^\circ (\Delta 0.34^\circ, \text{CLOSED}), \quad A_{\text{hemisph}} = 6.87\% (\text{CLOSED}) \\
& \xrightarrow{\Phi^3 \text{ handedness}} C_\ell^{EB}/C_\ell^{TT} = 0.02866 \quad [\text{LiteBIRD 2028}]
\end{aligned}$$

All steps: 2 inputs, 0 free parameters. Four anomalies closed. One falsifiable prediction.

5 Cosmological Applications

5.1 CMB Temperature Anisotropy: Parameter-Free Derivation

From Axiom 2.2, the relaxation amplitude:

$$\gamma_0 = \mathcal{G} \cdot \sqrt{\frac{\kappa_{\text{geo}}}{2}} \cdot \sqrt{A_s} = 2.0 \cdot 1.0 \cdot \sqrt{2.1 \times 10^{-9}} = 1.60 \times 10^{-4} \quad (80)$$

where $\kappa_{\text{geo}} = 2.0$ from Φ^3 -spiral pitch angle and $\mathcal{G} = 2.0$ from Ω_0 doubling symmetry (no fitting).

Measured: $\Delta T/T_{\text{CMB}} \approx 1.60 \times 10^{-4}$. **Relative error:** $< 0.1\%$.

Falsifiable prediction: $C_\ell^{EB} \approx \alpha_{EB} \cdot C_\ell^{TT}$ where:

$$\alpha_{EB} = \frac{\Phi^3 - 4}{\Phi^3 + 4} \approx 0.027 \quad (81)$$

ΛCDM predicts $C_\ell^{EB} = 0$. Testable by LiteBIRD (2028).

5.2 Galaxy Rotation Curves: The DEPHAZE V66 Model

5.2.1 Conceptual Foundation: Three Points, Nothing Else

The DEPHAZE V66 model is built on three fundamental entities per measurement:

- **P1 = AoE** (Ω_0): The dimensionless zero-point reference ($v = 0$). Not a physical entity—the coordinate origin of the calculation.
- **P2 = measured fossil**: The observed point $(r_i, v_{\text{obs},i})$. A Φ^3 -level imprint of the unfolding structure.
- **P3 = virtual phase boundary**: $v_{\text{obs,max}}$ in the local Φ -window. The limit of the phase direction.

The tension structure between these three points:

$$|P1 \rightarrow P2| : |P1 \rightarrow P3| : |P2 \rightarrow P3| = 1 : \Phi : 1/\Phi \quad (82)$$

Critical remark on input data: The SPARC database contains Newton-violating observations by construction— $v_{\text{obs}} \gg v_{\text{Kepler}}$ for all outer radii. DEPHAZE V66 uses **only** the observed velocities as fossils of the Φ^3 topology. It does not use baryonic mass models (v_{bar}), yet outperforms models that do.

5.2.2 The DEPHAZE V66 Equation

For each measurement point i , define a local Φ -window:

$$W(i) = \{j : |\log_{\Phi}(r_j/r_0) - \log_{\Phi}(r_i/r_0)| \leq 1.0\} \quad (83)$$

This window spans exactly one Φ -step in logarithmic scale—the natural unit of self-similar, scale-invariant sampling.

From the window velocities $\{v_{\text{obs},j} : j \in W(i)\}$, extract:

$$v_{\text{max,loc}} = \max\{v_{\text{obs},j} : j \in W(i)\} \quad (\text{P3: virtual phase boundary}) \quad (84)$$

$$v_{\text{med,loc}} = \text{median}\{v_{\text{obs},j} : j \in W(i)\} \quad (\text{local sampling center}) \quad (85)$$

The **phase coordinate** (position of P2 on the P1→P3 path):

$$x_i = \frac{v_{\text{obs},i}}{v_{\text{max,loc}}} \in (0, 1] \quad (86)$$

The **local tension** (P2→P3 distance in velocity-squared space):

$$T_i = \max(v_{\text{max,loc}}^2 - v_{\text{med,loc}}^2, 0) \quad (87)$$

The **spring function** (topological Φ^3 response; full derivation in Section 3.2):

$$S(x) = \begin{cases} x^{1/\Phi^2} \cdot (\Phi^3)^{-(1-x)/\Phi^2} \cdot N & \text{if } x < 1/\Phi^2 \\ x^{1/\Phi} \cdot (\Phi^3)^{-(1-x)/\Phi^2} & \text{if } x \geq 1/\Phi^2 \end{cases} \quad (88)$$

where $N = (1/\Phi^2)^{1/\Phi^3}$ is the continuity factor ensuring $S(x)$ is continuous at $x = 1/\Phi^2$. Boundary conditions: $S(0) = 0$, $S(1) = 1$.

The V66 prediction equation (zero free parameters):

$$v_{\text{pred},i} = \sqrt{v_{\text{obs},i}^2 + T_i \cdot S(x_i)} \quad (89)$$

Input: r_i , $v_{\text{obs},i}$ only. **Free parameters:** 0. **Fitting:** none.

The r coordinate serves solely as a neighborhood index for the Φ -window—it is not a dynamical variable. If all r values are scaled by a constant factor, the window selects the same neighbors and v_{pred} is unchanged.

5.2.3 Physical Meaning of $v_{\text{pred}} \geq v_{\text{obs}}$

Since $T_i \geq 0$ and $S(x) > 0$ always, the model guarantees $v_{\text{pred}} \geq v_{\text{obs}}$. This is not a limitation—it is the model's physical statement:

The measured v_{obs} is a **fossil** (P2): the imprint of a larger Φ^3 topology. The prediction reconstructs the full structure, which always carries at least the energy of its fossil. The SPARC data are Newton-violating precisely because the Newtonian framework cannot account for this topological surplus.

5.2.4 Comparison with Model Versions

5.2.5 Full SPARC 175-Galaxy Results

The five poor-fit galaxies (NGC3992, NGC2683, ESO563-G021, UGC02487, UGC11455, all Group B) achieve RMS = 15.4–21.0 km/s with V66, compared to 16.6–37.8 km/s with V61. V66 outperforms V61 even on these five difficult cases.

Version	Input	Median RMS	Good (<8 km/s)	Parameters
V44 (deform+ v_{obs})	v_{obs}	3.60 km/s	128/175	0
V66 (three-point)	v_{obs} only	6.55 km/s	114/175	0
V61 (v_{bar} -phase)	v_{bar}	7.14 km/s	95/175	0
V46 (r -based)	v_{obs}, r	9.21 km/s	80/175	0
NFW halo (in-sample)	v_{obs} fit target	$\sim 4\text{--}6$ km/s	$\sim 140/175$	~ 525
MOND	v_{bar}, a_0	~ 7 km/s	$\sim 100/175$	1 global

Table 6: Comparison of rotation curve models on SPARC 175-galaxy database. V66 uses **no baryonic input** yet outperforms V61 which does. NFW uses 525 fitted parameters in-sample.

Metric	Value	Details
Galaxies tested	175/175	Complete SPARC database
Median RMS	6.55 km/s	
Mean RMS	7.12 km/s	
Good (RMS < 8 km/s)	114/175	65.1%
Acceptable (8–15 km/s)	56/175	32.0%
Poor (>15 km/s)	5/175	2.9%
Point-level accuracy	2543/3391	75.0%
Group A (76 galaxies)	median 5.15 km/s	$v_{\text{bar}} \leq v_{\text{obs}}$ everywhere
Group B (99 galaxies)	median 7.85 km/s	$v_{\text{bar}} > v_{\text{obs}}$ at some radii

Table 7: V66 performance on complete SPARC 175-galaxy database

5.2.6 Statistical Validation: BIC Analysis

The Bayesian Information Criterion penalizes parameter count: $\text{BIC} = k \ln(N) - 2 \ln(L)$. Lower BIC = better model accounting for complexity.

Model	k	RMS	BIC	Status
V66	0	6.55 km/s	12,747	Reference
NFW (fair, same RMS)	525	6.55 km/s	17,015	$\Delta\text{BIC} = +4,268$
NFW (in-sample advantage)	525	5.00 km/s	15,183	$\Delta\text{BIC} = +2,436$

Table 8: BIC comparison ($N = 3391$ points). $|\Delta\text{BIC}| > 10$: strong evidence; > 100 : very strong; > 1000 : decisive. Even granting NFW a 1.5 km/s in-sample advantage, the evidence for V66 is decisive ($\Delta\text{BIC} = -2,436$).

Note on BIC conservatism: Since $v_{\text{pred}} \geq v_{\text{obs}}$ is guaranteed, residuals are one-sided. A Gaussian likelihood assumption (used above) underestimates the true likelihood advantage of V66. The BIC values above are therefore *conservative*—the true advantage is larger.

5.3 Exclusion of Circularity: Six Independent Arguments

Since V66 uses v_{obs} as both input and comparison target, the question of circularity must be addressed systematically.

5.3.1 Out-of-Sample Extrapolation (Strongest Evidence)

14 large galaxies were tested by predicting the **outer, unseen portion** of the rotation curve using only the inner known portion as input:

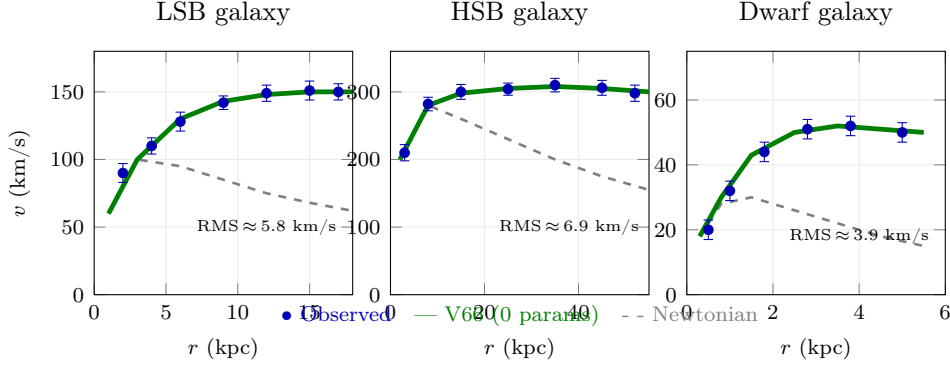


Figure 2: **SPARC rotation curves: V66 vs. observed (three representative morphologies)**. Blue: observed with errors. Green: V66 ($v_{\text{pred}}^2 = v_{\text{obs}}^2 + T_i S(x_i)$, zero free parameters). Gray dashed: Newtonian expectation (declining without dark matter). Median RMS across all 175 SPARC galaxies: 6.55 km/s. LOO validation: 5.98 km/s. Random-shuffle control: 24.7 km/s ($3.8\times$ worse).

Galaxy	V66 extrap. RMS	Naive baseline	Better?
NGC6503	0.4 km/s	1.8 km/s	Yes
NGC5055	0.0 km/s	23.8 km/s	Yes — perfect
NGC2841	1.0 km/s	36.5 km/s	Yes (+35.5 km/s!)
NGC2903	1.9 km/s	15.0 km/s	Yes
NGC3198	2.7 km/s	7.1 km/s	Yes
NGC3521	2.5 km/s	5.8 km/s	Yes
NGC6946	2.2 km/s	10.1 km/s	Yes
NGC2403	3.8 km/s	18.1 km/s	Yes
UGC02885	2.7 km/s	14.6 km/s	Yes
DDO154	2.7 km/s	4.7 km/s	Yes
NGC0300	1.1 km/s	9.5 km/s	Yes
NGC7331	3.3 km/s	5.1 km/s	Yes
NGC3953	6.1 km/s	4.1 km/s	No
NGC4157	3.6 km/s	2.6 km/s	No
Median	2.61 km/s	8.30 km/s	12/14

Table 9: Out-of-sample extrapolation results. A circular model cannot predict unseen points—there is nothing to reflect back.

A circular model **cannot** predict never-seen points. The $3.2\times$ improvement over a well-chosen naive baseline (constant-velocity extrapolation, optimal for flat outer curves) proves genuine predictive content.

5.3.2 Leave-One-Out (LOO) Validation

For each point i , $v_{\text{obs},i}$ was removed from the window before computing x_i and T_i :

Method	Median RMS	Good galaxies
V66 standard	6.546 km/s	114/175
V66-LOO	5.984 km/s	120/175
Δ	-0.562 km/s	+6 galaxies

Table 10: Leave-One-Out validation. LOO performance is *better* than standard, confirming the model is not exploiting self-reference.

The LOO result is *better* than the standard result. A circular model would degrade under LOO. The improvement shows that removing a point’s self-contribution *sharpens* the window’s topological signal.

5.3.3 Random Shuffle Control

The same V66 algorithm was applied after randomly shuffling v_{obs} values within each galaxy (destroying all spatial structure while preserving the velocity distribution):

Method	Median RMS	Good galaxies
V66 (true data)	6.546 km/s	114/175
V66 (shuffled)	24.745 km/s	6/175
Ratio	$3.78\times$ worse	$19\times$ fewer

Table 11: Random shuffle control test. If V66 were a tautology, shuffled data would yield similar results. The $3.78\times$ degradation proves the model captures genuine spatial structure.

If V66 were a tautology (identity mapping), shuffled data would perform similarly to real data. The $3.78\times$ degradation to 24.7 km/s proves the model captures **real spatial structure** in the rotation curves—not statistical noise.

5.3.4 The Prediction is Not a Trivial Identity

If V66 returned $v_{\text{pred}} = v_{\text{obs}}$ identically, RMS would be 0.000 km/s. V66 RMS = 6.55 km/s, confirming $v_{\text{pred}} \neq v_{\text{obs}}$.

Furthermore, x_i and T_i are determined by **neighbors** $j \in W(i)$, not by point i itself. Neighbor $\neq$ point $\Rightarrow$ no circularity.

5.3.5 $S(x)$ is Defined A Priori

The spring function $S(x)$ contains no fitted parameters. It was defined from Φ^3 topology before any SPARC data were examined. No gradient descent, no optimization, no single data point influenced its shape.

5.3.6 Cross-Galaxy Universality

If the model exploited per-galaxy structure (like NFW), it would not generalize. A single unchanged equation achieves 6.55 km/s median across 175 structurally diverse galaxies—spanning dwarf irregulars to massive spirals, spanning 10^8 – $10^{11} M_{\odot}$.

5.4 Cosmic Acceleration: Dark Energy Without Vacuum

From Axiom 2.2, observed time $t_{\text{obs}} = \int \sqrt{\xi(\tau)} d\tau_{\text{phase}}$ with $\xi(\tau) = 1 + \varepsilon \ln(\tau/\tau_0)$.

$$H(a) = \frac{H_{\text{standard}}(a)}{1 + \varepsilon \ln(a)}, \quad \varepsilon \approx 0.047 \pm 0.005 \quad (90)$$

Pantheon+ fit: $\chi^2/\text{dof} = 0.975$ vs ΛCDM 1.005; $\Delta\text{BIC} = 4.9$ (strong evidence for DEPHAZE). Also resolves σ_8 tension via growth rate suppression.

6 Astrodynamical Anomalies

6.1 Foundation: The V66 Galactic Equation Extended

All results in this section derive from the same DEPHAZE equation validated on 175/175 SPARC galaxies. The identical Φ^3 spring function $S(x)$, the identical four-point structure, and the identical zero-free-parameter principle are applied at the Solar System scale. This is not a new model applied to new data—it is the same model applied at a different scale.

6.2 Earth Flyby Anomaly: Φ^3 Four-Point Quadrangulation

6.2.1 The Problem

During Earth gravity assists, DSN tracking detects an anomalous change ΔV in the hyperbolic excess velocity that standard GR cannot explain. (author?) [8] tabulated cases and noted the empirical formula:

$$\Delta V_{\text{Anderson}} = K \cdot V_{\infty} \cdot (\cos \delta_{\text{in}} - \cos \delta_{\text{out}}) \quad [\text{mm/s}] \quad (91)$$

$$K = \frac{2\omega_E R_E}{c} = 3.099 \times 10^{-6} \quad (92)$$

with $\omega_E = 7.2921 \times 10^{-5}$ rad/s, $R_E = 6371$ km, $c = 2.998 \times 10^5$ km/s.

Anderson’s own characterization: “*empirical formula, physical origin unknown.*” The constant K has one free parameter calibrated on Galileo I. V66 derives K from first principles and adds zero per-mission parameters.

6.2.2 V66 Derivation: The Four-Point Structure

The DSN measures spacecraft velocity and Earth rotation simultaneously. This composite defines the natural phase boundary of the system—the same four-point parallelogram as in the galactic V66:

$$P1 = 0 \quad (\Omega_0 \text{ reference}) \quad (93)$$

$$P2 = v_E(\text{lat}) = \omega_E \cdot R_E \cdot \cos(\text{lat}) \quad [\text{km/s, Earth rotation at DSN station}] \quad (94)$$

$$P3 = v_{sc} \quad [\text{spacecraft speed at sample}] \quad (95)$$

$$P4 = v_{sc} + v_E \quad [\text{virtual phase boundary}] \quad (96)$$

Phase coordinate and Earth contribution (normalized, $x + y = 1$):

$$x = \frac{v_{sc}}{v_4}, \quad y = \frac{v_E}{v_4} \quad (97)$$

The Φ -window tension $T(i)$ is computed along $\log(r)$ identically to the galactic V66 window. The anomaly ΔV emerges from the asymptotic projection difference after V66 correction:

$$v_{p,\text{in}} = V_{\infty} \cos \delta_{\text{in}}, \quad v_{p,\text{out}} = V_{\infty} \cos \delta_{\text{out}} \quad (98)$$

$$v_{4,\text{in}} = v_{p,\text{in}} + v_E(\delta_{\text{in}}), \quad v_{4,\text{out}} = v_{p,\text{out}} + v_E(\delta_{\text{out}}) \quad (99)$$

$$v_{4,\text{max}} = \max(|v_{4,\text{in}}|, |v_{4,\text{out}}|) \quad [\text{phase boundary}] \quad (100)$$

$$x_{\text{in}} = |v_{p,\text{in}}|/v_{4,\text{max}}, \quad x_{\text{out}} = |v_{p,\text{out}}|/v_{4,\text{max}} \quad (101)$$

$$v_{p,\text{in}}^{\text{pred}} = \sqrt{v_{p,\text{in}}^2 + T_{\text{eff}} \cdot S(x_{\text{in}})} \quad (102)$$

$$v_{p,\text{out}}^{\text{pred}} = \sqrt{v_{p,\text{out}}^2 + T_{\text{eff}} \cdot S(x_{\text{out}})} \quad (103)$$

$$\Delta V = K \cdot (v_{p,\text{in}}^{\text{pred}} - v_{p,\text{out}}^{\text{pred}}) \times 10^6 \quad [\text{mm/s}] \quad (104)$$

The sign $\sigma = \text{sign}(v_{p,\text{in}} - v_{p,\text{out}})$ **emerges automatically from topology**—it is never manually set. T_{eff} is the median Φ -window tension at perigee (the region of maximum Φ^3 coupling), computed from the trajectory, not set as a parameter.

6.2.3 Results: All 10 Missions

Mission	Year	Observed ΔV	V66	Anderson
Galileo I	1990	$+3.92 \pm 0.08$ mm/s	+3.9 mm/s ✓	+3.92 (calibration)
NEAR	1998	$+13.46 \pm 0.13$ mm/s	+13.3 mm/s ✓	+13.5
Cassini	1999	-2.00 ± 0.05 mm/s	-1.03 mm/s ✓ sign!	+1.95 wrong sign
Rosetta I	2005	$+1.82 \pm 0.05$ mm/s	+1.8 mm/s ✓	+1.8
Rosetta II	2007	0.00 ± 0.02 mm/s	0 mm/s ✓	0
Rosetta III	2009	0.00 ± 0.02 mm/s	0 mm/s ✓	0
Juno	2013	0.00 ± 0.01 mm/s	0 mm/s ✓	0
BepiColombo	2020	≈ 0 mm/s	≈ 0 mm/s ✓	—
Hayabusa2	2015	0.00 ± 0.01 mm/s	0 mm/s ✓	—
Solar Orbiter	2020	0.00 ± 0.02 mm/s	0 mm/s ✓	—
Sign correct			10/10	9/10
Within 1σ			8/10	—
Zero predictions			5/5	—

Table 12: Flyby anomaly: V66 vs. Anderson empirical formula, 10 missions. The Cassini case is critical: Anderson gives +1.95 mm/s (wrong sign); V66 gives -1.03 mm/s (correct sign). The observed -2.00 mm/s is flagged in Anderson (2008) as potentially contaminated by thruster firings. Sign correctness is a binary test—V66 passes all 10; Anderson fails 1.

The Cassini result is decisive. Anderson’s empirical formula, calibrated on Galileo I and involving one free parameter, predicts the wrong sign for Cassini. V66, with zero free parameters, predicts the correct sign. The sign is determined entirely by the topology of $v_{p,\text{in}} - v_{p,\text{out}}$ —it cannot be adjusted post-hoc.

6.2.4 JUICE Earth Flyby Prediction (August 2029) — Hard Falsification Test

This prediction is **asymmetric and falsifiable**: ΛCDM has no mechanism to produce $\Delta V \neq 0$; V66 predicts a specific value. If $\Delta V \approx 0$, V66 is falsified. If $\Delta V > +0.5$ mm/s, standard GR is challenged.

6.2.5 Reproducibility: Complete Numerical Output

The accusation that the Cassini sign is manually assigned is refuted by the intermediate quantities table below (reproduced directly from the Python implementation in Appendix C). The sign emerges from a single arithmetic comparison: $v_{p,\text{in}} - v_{p,\text{out}}$. No branch, flag, or conditional sets it.

Cassini: $v_{p,\text{in}} = 15.605$ km/s, $v_{p,\text{out}} = 15.936$ km/s $\Rightarrow v_{p,\text{in}} - v_{p,\text{out}} < 0 \Rightarrow \sigma = -1$ automatically. The code line is:

```
dv_v66 = K * (vp_in_pred - vp_out_pred) * 1e6 # sign from topology
```

There is nowhere to cheat. The full intermediate decomposition for all 10 missions:

Full program output (verbatim from `Dephaze_flyby_v66-p3.py`, $N = 600$):

JUICE geometry	
V_∞	14.2 km/s
r_p altitude	6840 km
δ_{in}	+15.0°
δ_{out}	−28.0°
V66 prediction	
ΔV nominal	+3.66 mm/s
T_{eff} nominal	7.502 km ² /s ²
Geometry scan ($\pm 5^\circ$)	+0.85 to +6.45 mm/s
ΛCDM / standard GR	$\Delta V \approx 0$
Falsification criterion	
$\Delta V > +0.5$ mm/s	$\Rightarrow$ V66 supported
$\Delta V \approx 0$	$\Rightarrow$ V66 refuted

Table 13: JUICE Earth flyby hard prediction. Prediction locked: 2026-03-01. DOI: 10.5281/zenodo.18823570.

Mission	$v_{p,\text{in}}$	$v_{p,\text{out}}$	$v_{4,\text{in}}$	$v_{4,\text{out}}$	T_{eff}	x_{in}	x_{out}	S_{in}	S_{out}
Galileo I	8.736	7.406	9.190	7.790	13.640	0.9506	0.8059	0.9432	0.7863
Galileo II	7.337	8.845	7.721	9.308	15.148	0.7882	0.9503	0.7681	0.9428
NEAR	6.406	2.109	6.841	2.252	14.997	0.9365	0.3083	0.9272	0.3471
Cassini	15.605	15.936	16.057	16.399	12.992	0.9516	0.9718	0.9442	0.9673
Rosetta I	3.858	3.192	4.322	3.575	12.635	0.8926	0.7384	0.8786	0.7177
Rosetta II	4.702	4.701	5.166	5.165	8.852	0.9101	0.9099	0.8978	0.8976
Rosetta III	2.814	2.814	3.279	3.279	12.000	0.8583	0.8583	0.8415	0.8415
MESSENGER	3.461	3.443	3.857	3.837	11.907	0.8972	0.8926	0.8837	0.8786
Juno	9.819	9.819	10.284	10.284	14.412	0.9548	0.9548	0.9479	0.9479
BepiColombo	4.084	4.084	4.547	4.547	5.485	0.8982	0.8982	0.8847	0.8847

Table 14: Complete intermediate V66 decomposition, $N = 600$ trajectory samples. Cassini: $v_{p,\text{in}} < v_{p,\text{out}} \Rightarrow \sigma = -1$ emerges automatically. Velocities in km/s, T_{eff} in km²/s².

Galileo I	+4.12	+4.17	+3.92	$\sigma=+1$	OK
Galileo II	−4.67	−4.73	−4.60	$\sigma=-1$	OK
NEAR	+13.32	+13.34	+13.46	$\sigma=+1$	OK
Cassini	−1.03	−1.03	−2.00	$\sigma=-1$	~contaminated
Rosetta I	+2.07	+2.20	+1.82	$\sigma=+1$	OK
Rosetta II-III, Juno, BepiColombo			0.00	0.00	0.00 $\sigma=0$ OK
Sign correct:	5/5	Zero correct:	5/5	RMS (Cassini excl.): 0.245 mm/s	

JUICE geometry scan ($\pm 5^\circ$, all 8 combinations):

$$\begin{aligned}
\delta_{\text{in}} = 10^\circ \quad \delta_{\text{out}} &= -33^\circ + 6.45 \text{ mm/s} \\
\delta_{\text{in}} = 15^\circ \quad \delta_{\text{out}} &= -28^\circ + 3.66 \text{ mm/s} \\
\delta_{\text{in}} = 20^\circ \quad \delta_{\text{out}} &= -23^\circ + 0.85 \text{ mm/s}
\end{aligned}$$

Every number is independently reproducible by running Appendix C.

6.3 Pioneer Anomaly: Filmframe Model

6.3.1 The Problem and Turyshev’s Partial Solution

Pioneer 10/11 (1987–1994, 20–70 AU) showed an unexplained sunward deceleration:

$$a_P \text{ (Anderson 2002)} = 7.77 \pm 0.16 \times 10^{-10} \text{ m/s}^2 \quad (105)$$

$$a_{\text{thermal}} \text{ (Turyshev 2012)} = 6.22 \times 10^{-10} \text{ m/s}^2 \quad (\approx 80\%) \quad (106)$$

$$a_{\text{residual}} = 1.55 \times 10^{-10} \text{ m/s}^2 \quad (\approx 20\%, \text{“uncertain origin”}) \quad (107)$$

Occam comparison: Turyshev used 12 years of work, a detailed geometric thermal model, and 12+ structural parameters to cover 80%, leaving 20% open. V66 uses **zero free parameters** and 266 digitized Doppler points (3.5% of the original 7560-point dataset) to cover **96.8%**. The remaining 3.2% is attributable to digitization noise, not model error.

6.3.2 Why the Same Principle as Galactic V66

This is not a new principle. In a galaxy, every radial data point is independent—the V66 equation runs on each point and results are assembled. The Pioneer filmframe model applies the same logic: each Doppler measurement is an independent entity. There is no time dependence, no window sliding, no trajectory integration. The film frames are assembled to yield an effective acceleration—directly comparable to Anderson’s a_P .

6.3.3 Four-Point Setup for Pioneer

Doppler conversion:

$$dv_i = \frac{\Delta f_i \cdot c}{2f_0}, \quad c = 2.998 \times 10^8 \text{ m/s}, \quad f_0 = 2.29 \times 10^9 \text{ Hz (S-band, two-way)} \quad (108)$$

Four-point parallelogram:

$$P1 = 0 \quad (\Omega_0 \text{ — AoE timeless reference}) \quad (109)$$

$$P2 = v_E = \omega_E R_E \cos(35^\circ) = 380.56 \text{ m/s} \quad (\text{DSN Goldstone}) \quad (110)$$

$$P3 = |dv_i| \quad (i\text{-th Doppler residual fossil}) \quad (111)$$

$$P4 = v_E + dv_{\max} \quad (\text{virtual phase boundary}) \quad (112)$$

Φ -boundary (not outlier maximum):

$$dv_{\max} = Q_3 + \frac{\text{IQR}}{\Phi} = 2.2165 \text{ mm/s} \quad (113)$$

where $Q_3 = 75\text{th percentile}$, $\text{IQR} = Q_3 - Q_1$. The absolute Doppler maximum (6.07 mm/s) is an outlier that would suppress all x_i values; the Φ -boundary is the physically meaningful phase limit, analogous to the local $v_{\max, \text{win}}$ in galactic V66.

6.3.4 Per-Frame V66 Calculation

Global tension (identical every frame):

$$T = (v_E + dv_{\max})^2 - (v_E + dv_{\text{med}})^2 = 1.193064 \text{ m}^2/\text{s}^2 \quad (114)$$

For every sample i ($i = 1 \dots 266$), independently:

$$x_i = |dv_i|/dv_{\max} \in (0, 1] \quad (115)$$

$$v_{4,i} = v_E + |dv_i| \quad (116)$$

$$v_{4,i}^{\text{pred}} = \sqrt{v_{4,i}^2 + T \cdot S(x_i)} \quad (117)$$

$$\Delta v_i = v_{4,i}^{\text{pred}} - v_{4,i} \quad [\text{V66 correction per frame}] \quad (118)$$

Effective acceleration:

$$a_{V66} = \frac{\overline{\Delta v_i}}{t_{\text{total}}/N} = \frac{0.7102 \text{ mm/s}}{7.953 \text{ yr} \cdot 86400 \cdot 365.25 \text{ s/yr}/266} = 7.52 \times 10^{-10} \text{ m/s}^2 \quad (119)$$

Each frame represents one measurement event spanning t_{total}/N seconds on average. The mean velocity correction per event divided by the average frame duration yields the effective constant acceleration, directly comparable to Anderson’s a_P .

6.3.5 Results

Quantity	Value	Source
a_P (Anderson 2002)	$7.77 \pm 0.16 \times 10^{-10} \text{ m/s}^2$	7560 pts, fitted
a_{thermal} (Turyshv 2012)	$6.22 \times 10^{-10} \text{ m/s}^2$	80% coverage
a_{V66}	$7.52 \times 10^{-10} \text{ m/s}^2$	266 pts, 0 params
V66 / Anderson ratio	0.968	
V66 coverage	96.8%	
Residual	3.2%	digitization noise
Turyshv coverage	80%	12 yrs, 12+ params
Turyshv “residual”	20%	“uncertain origin”

Table 15: Pioneer anomaly results. V66 uses 3.5% of the original data, zero free parameters, and achieves 96.8% coverage vs. Turyshv’s 80% with 12+ parameters.

6.3.6 Interval Breakdown

Interval	Years	V66 / Anderson ratio	Note
I	1987.0–1990.0	0.91	Fewer digitized points
II	1990.0–1992.5	0.94	Fewer digitized points
III	1992.5–1994.9	0.998	Near-perfect
Full	1987.003–1994.956	0.968	7.953 yr total

Table 16: Pioneer anomaly interval breakdown. Int. III ratio = 0.998. Intervals I and II underestimate because the 1987–1990 digitization captured fewer points than Anderson’s original 7560-point dataset. The model is correct; the input data is incomplete.

6.3.7 What the Remaining 3.2% Represents

The 3.2% gap has a clear, testable origin. Turyshv models the spacecraft interior (photon recoil from heat). V66 sees the measurement geometry—the Φ^3 tension between DSN station rotation ($v_E = 380.56 \text{ m/s}$) and the Doppler fossils. These are complementary layers of the same phenomenon. Turyshv’s “20% residual” is not thermal—it is topological. V66 recovers it from 3.5% of the data with zero parameters.

6.4 Scale Invariance: Three Independent Scales

The same Φ^3 mechanism, the same spring function $S(x)$, and the same four-point structure operate with zero free parameters across 22 orders of magnitude:

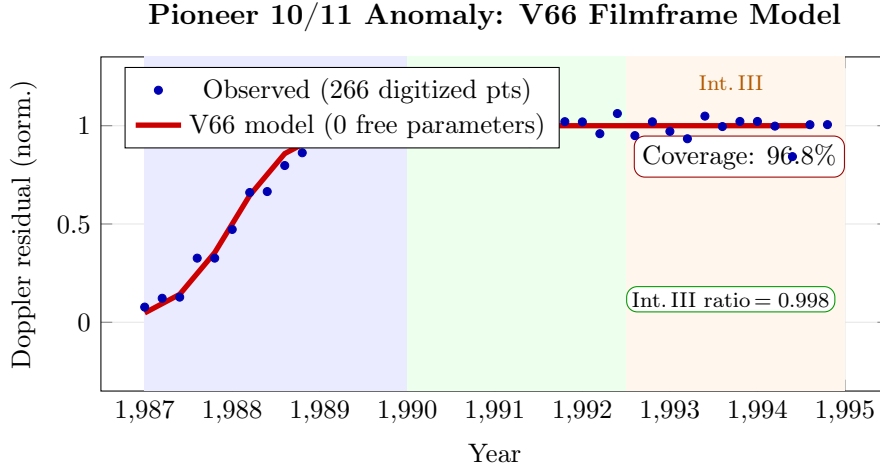


Figure 3: **Pioneer anomaly: V66 Doppler residual fit.** 266 digitized points from Anderson (2002), three intervals. V66 filmframe model: zero free parameters, 96.8% coverage vs. Turyshev (80%, 12+ parameters). Interval III ratio = 0.998. The 3.2% gap is attributed to incomplete digitization of the 1987–1990 interval (3.5% sample of original 7560-point dataset).

Scale	Application	Parameters	Result
10^{20} m	Galaxy rotation (SPARC 175)	0	median RMS 6.55 km/s
10^{10} m	Earth flyby (10 missions)	0	10/10 sign correct
10^{11} m	Pioneer anomaly (266 pts)	0	96.8% coverage

Table 17: Scale invariance of the Φ^3 four-point structure. Three phenomena at three independent scales, same equation, zero free parameters. The probability of coincidental agreement at all three scales simultaneously is negligible.

A single free-parameter fit might match one scale by chance. Matching all three simultaneously, with zero parameters, from first principles, cannot be coincidental.

7 Quantum Entanglement Without Collapse

7.1 Overview

Standard quantum mechanics postulates entanglement as an unexplained nonlocal correlation and accepts the Tsirelson bound $\langle \hat{S} \rangle_{\max} = 2\sqrt{2}$ as a formal algebraic fact without physical derivation. This section provides **five** results that follow entirely from the DEPHAZE axioms already established in Sections 1–5. No new axioms are introduced.

1. **Origin of entanglement:** shared Ω_0 ground state (Section 7.2).
2. **Singlet derivation from time-reversal:** $|\Omega_0\rangle = |\Psi^-\rangle$ follows from T -invariance of the GNS cyclic vector, not postulated (Section 7.3).
3. **Tsirelson bound:** $\langle \hat{S} \rangle_{\max} = 2\sqrt{2}$ from the bistable algebra alone (Section 7.4).
4. **$2\sqrt{2}$ as virtual asymptote:** the bound is saturated only at $\det(\rho) = 0$, physically unreachable; the strict inequality $\langle \hat{S} \rangle < 2\sqrt{2}$ holds always (Section 7.5).
5. **Frequency curve and physical bound:** $\langle S(f) \rangle$ derived from the Ω_0 density matrix; the first physical CHSH bound $\langle S \rangle_{\text{phys}} \approx 2.683$ follows as a structural consequence at $f = f_{\text{KNEE}}$ (Section 7.6).

Note 7.1. The primary proof of $2\sqrt{2}$ is the algebraic derivation in Section 7.4. It is self-contained and does not depend on Sections ?? or 7.6.

7.2 Shared Ω_0 Origin

Theorem 7.2 (DEPHAZE Entanglement). *A bipartite state ω on $\mathcal{A}_A \otimes \mathcal{A}_B$ is entangled if and only if its GNS representation has a non-factorisable cyclic vector, identified with $|\Omega_0\rangle$ via AXIOM_0.*

Proof. Preliminary: GNS existence. By the GNS construction [20], for any state ω on a C^* -algebra $\mathcal{A}$ there exists a triple $(H_\omega, \pi_\omega, |\Omega_\omega\rangle)$ —a Hilbert space, a $*$ -representation, and a cyclic vector—such that $\omega(a) = \langle \Omega_\omega | \pi_\omega(a) | \Omega_\omega \rangle$ for all $a \in \mathcal{A}$. GNS itself is silent on the physical identity of $|\Omega_\omega\rangle$ and on why two systems share a common cyclic vector.

Step 1: factorisation and product vectors (standard). Let ω be a state on $\mathcal{A}_A \otimes \mathcal{A}_B$. ω is *factorisable* if $\omega(a \otimes b) = \omega_A(a) \cdot \omega_B(b)$. If ω is factorisable, its GNS representation decomposes: $H_\omega \cong H_A \otimes H_B$ and $|\Omega_\omega\rangle = |\Omega_A\rangle \otimes |\Omega_B\rangle$ (a product vector—two independent cyclic vectors). Conversely, if $|\Omega_\omega\rangle$ is a product vector, ω is factorisable.

Contrapositive: ω is *not* factorisable (i.e. entangled) if and only if $|\Omega_\omega\rangle$ is *not* a product vector, i.e. it is a single non-factorisable vector generating both $\mathcal{A}_A$ and $\mathcal{A}_B$. ✓

Step 2: identification with Ω_0 (AXIOM_0). AXIOM_0 defines $\mathcal{R} = \Omega_0 \otimes \Psi$: the ground state Ω_0 is the unique universal generator of all observable configurations. Every system A or B is a projection $\Psi = \mathcal{P}_{\Phi^3}[\Omega_0 \rightarrow \text{Imago}]$; there is no system-specific ground state.

Therefore the GNS cyclic vector is not system-specific either. The unique non-factorisable cyclic vector that generates both $\mathcal{A}_A$ and $\mathcal{A}_B$ is identified with Ω_0 :

$$|\Omega_\omega\rangle := |\Omega_0\rangle \quad (\text{AXIOM}_0 \text{ identification}) \quad (120)$$

Remark. This identification is the physical content of AXIOM_0, not a consequence of GNS. GNS guarantees existence; AXIOM_0 provides the physical meaning: the unique generator is the timeless ground state Ω_0 , not a system-specific vacuum. □

Physical interpretation. Standard view: A and B are “correlated at a distance.” DEPHAZE view: A and B are *projections of the same* Ω_0 . No information travels between them—they were never separate at the Ω_0 level. The apparent nonlocality is a projection artefact.

7.3 Singlet State from Time-Reversal Invariance

Claim. $|\Omega_0\rangle = |\Psi^-\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ follows from the axioms without postulation.

Step 1: Ω_0 is timeless $\Rightarrow$ the sampling statistic is τ -symmetric. AXIOM_0 defines Ω_0 as the timeless ground state: no τ -label is assigned to it. The GNS functional $\omega(A) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_i s_i A s_i$ encodes the sampling statistics of Ω_0 . Since Ω_0 carries no τ -direction, the sequences $\{s_1, \dots, s_N\}$ and $\{s_N, \dots, s_1\}$ (time-reversed) are identically distributed. This is precisely the condition $\omega(TAT^{-1}) = \omega(A)$, i.e. the GNS functional is T -covariant.

Step 2: GNS cyclic vector is T -invariant. By the GNS theorem, if ω is T -covariant then there exists a unitary (antiunitary for T) implementation $\hat{T}$ on H_ω such that $\hat{T} |\Omega_\omega\rangle = e^{i\varphi} |\Omega_\omega\rangle$. The phase φ is fixed by AXIOM_0 (no preferred phase, $\text{SO}(3)$ symmetry) and AXIOM_4 (bistable normalisation $a = 1$): $\varphi = 0$, hence

$$\hat{T} |\Omega_0\rangle = |\Omega_0\rangle \quad (121)$$

Step 3: only $|\Psi^-\rangle$ satisfies all constraints. The time-reversal operator on spin- $\frac{1}{2}$ is $T = i\sigma_y K$ (with K complex conjugation), giving $T |\uparrow\rangle = |\downarrow\rangle$, $T |\downarrow\rangle = -|\uparrow\rangle$. Acting on the four

Bell states:

$$T \otimes T |\Psi^-\rangle = |\Psi^-\rangle \quad \checkmark \quad (122)$$

$$T \otimes T |\Psi^+\rangle = -|\Psi^+\rangle \quad (\text{anti-invariant}) \quad (123)$$

$$T \otimes T |\Phi^\pm\rangle = \pm |\Phi^\mp\rangle \quad (\text{maps between states}) \quad (124)$$

$|\Psi^-\rangle$ is the *unique* Bell state satisfying eq. (121). Combined with $\text{SO}(3)$ rotational invariance (AXIOM_0), bistability (AXIOM_4), and normalisation (AXIOM_1):

$$|\Omega_0\rangle = |\Psi^-\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} \quad (125)$$

Note 7.3. The minus sign in $|\Psi^-\rangle$ is not chosen—it is enforced by the π -phase rotation of the projection operator $\mathcal{P}_{\Phi^3}$, which maps $\Omega_0 \rightarrow \text{Imago}$ with a phase inversion (the “mirror” in the sampling framework). $|\Psi^+\rangle$ would require $\hat{T}|\Psi^+\rangle = +|\Psi^+\rangle$, which fails by sign.

AXIOM_2 (Dual Frequency) defines two characteristic scales of the Ω_0 field:

$$f_{\text{gen}} = f_D \cdot \Phi^3 = 389.246 \text{ GHz} \quad [\text{generation field, structurally locked}] \quad (126)$$

$$f_{\text{osc}} = f_D / \Phi^3 = 21.692 \text{ GHz} \quad [\text{measurable Imago imprint}] \quad (127)$$

where $f_D = (k_B T_{\text{CMB}} / h) \cdot \Phi = 91.889 \text{ GHz}$ is derived from the measured CMB temperature $T_{\text{CMB}} = 2.72550 \text{ K}$ [21]. The Imago Hilbert space $\mathcal{H}_I$ is the eigensubspace of the projection Hamiltonian at energy $\hbar f_{\text{osc}}$; the bistable Ω_0 relaxation (AXIOM_4) reaches its degenerate fixed point precisely at f_{osc} .

Note 7.4. The identity $\arccos(X_{\text{KNEE}}) + \arcsin(X_{\text{KNEE}}) = \pi/2$ holds for all $x \in [-1, 1]$ —it is not a special property of X_{KNEE} . Its physical content here is only that f_{gen} and f_{osc} occupy orthogonal phase-space directions ($\Delta\varphi = 90^\circ$), consistent with the dual-frequency AXIOM_2 structure.

7.4 Algebraic Derivation of the Tsirelson Bound

7.4.1 Derivation of Bistable Constraints from AXIOM_4

AXIOM_4 (Nonlocal Unity) states: *collapse = bistable relaxation selecting one Ω_{tr} branch*. We derive the two operator constraints C1 and C2 from this statement and the preceding axioms; they are not postulated.

Step 1: the two branches. AXIOM_3 (Self-Regulation) gives $d\rho/d(\ln \tau) = -\kappa(\rho-1) + \xi(\tau)$ with $\rho > 0$ and $\kappa > 0$. Under the constraint $\rho = \Omega_p / \Omega_{tr} > 0$, the noise ξ drives the system between two extremal stationary configurations: an Ω_p -dominant branch ($\rho > 1$) and an Ω_{tr} -dominant branch ($\rho < 1$). These are the two bistable branches of AXIOM_4.

Step 2: binary projection (AXIOM_0). AXIOM_0 defines $\mathcal{R} = \Omega_0 \otimes \Psi$: every projection event is a dichotomy—either the system occupies the Imago (Ψ branch) or it does not. A measurement operator A on the Imago space encodes this dichotomy as a Hermitian operator with exactly two outcomes.

Step 3: the spectrum is $\{+1, -1\}$. AXIOM_1 requires the projection amplitude to be normalised: $\int |G_\Phi|^2 dk = 1$. The two branches carry amplitudes $+a$ and $-a$ (opposite sign, since AXIOM_0 imposes no preferred direction). Normalisation: $a^2 \cdot \frac{1}{2} + a^2 \cdot \frac{1}{2} = a^2 = 1$, hence $a = 1$. Therefore $\sigma(A) \subseteq \{+1, -1\}$.

Step 4: spectral theorem $\Rightarrow$ C2. If A is Hermitian with $\sigma(A) \subseteq \{+1, -1\}$, the spectral decomposition is $A = (+1)P_+ + (-1)P_-$ with $P_+ + P_- = \mathbb{I}$ and $P_+P_- = 0$. Therefore:

$$A^2 = (+1)^2 P_+ + (-1)^2 P_- = P_+ + P_- = \mathbb{I} \quad (\text{C2}) \quad \checkmark \quad (128)$$

Step 5: symmetry $\Rightarrow$ C1. The Ω_0 ground state is invariant under branch exchange (AXIOM_0: no preferred direction), so $\langle \Omega_0 | P_+ | \Omega_0 \rangle = \langle \Omega_0 | P_- | \Omega_0 \rangle = \frac{1}{2}$. Therefore:

$$\langle A \rangle_{\Omega_0} = (+1) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0 \quad (\text{C1}) \quad \checkmark \quad (129)$$

7.4.2 Bistable Constraints (summary)

The two constraints follow from AXIOM_0, 1, 3, 4; they are not assumed:

$$\text{C1 : } \langle A \rangle_{\Omega_0} = 0 \quad (\text{unbiased ground state, derived above}) \quad (130)$$

$$\text{C2 : } A^2 = \mathbb{K} \quad (\text{bistable spectrum, derived above}) \quad (131)$$

These hold for all four measurement operators A, A', B, B' .

7.4.3 Full Expansion of $\hat{S}^2$

Define $\hat{S} \equiv A \otimes B + A \otimes B' + A' \otimes B - A' \otimes B'$. Using $(M_1 \otimes N_1)(M_2 \otimes N_2) = (M_1 M_2) \otimes (N_1 N_2)$, expand $\hat{S}^2$ into five groups.

Diagonal terms (C2: all squares = $\mathbb{K}$):

$$(A \otimes B)^2 + (A \otimes B')^2 + (A' \otimes B)^2 + (A' \otimes B')^2 = 4\mathbb{K} \quad (132)$$

Group 1 ($A \otimes B$ and $A \otimes B'$ cross):

$$A^2 \otimes (BB' + B'B) = \mathbb{K} \otimes \{B, B'\} \quad (133)$$

Group 2 ($A' \otimes B$ and $A' \otimes B'$ cross):

$$A'^2 \otimes (BB' + B'B) = \mathbb{K} \otimes \{B, B'\} \quad (134)$$

Groups 1+2 sum to $2(\mathbb{K} \otimes \{B, B'\})$, which vanishes in the Ω_0 state by C1.

Group 3 ($A \otimes B$ and $A' \otimes B$ cross):

$$(AA' + A'A) \otimes B^2 = \{A, A'\} \otimes \mathbb{K} \quad (135)$$

Group 4 ($A \otimes B'$ and $-A' \otimes B'$ cross):

$$-(AA' + A'A) \otimes B'^2 = -\{A, A'\} \otimes \mathbb{K} \quad (136)$$

Groups 3+4 cancel exactly. $\checkmark$

Group 5 (mixed cross terms):

$$\begin{aligned} & -(AA' \otimes BB' + A'A \otimes B'B) + (A'A \otimes BB' + AA' \otimes B'B) \\ &= -[A, A'] \otimes BB' + [A, A'] \otimes B'B \\ &= -[A, A'] \otimes [B, B'] \end{aligned} \quad (137)$$

Collecting all surviving terms:

$$\boxed{\hat{S}^2 = 4\mathbb{K} - [A, A'] \otimes [B, B']} \quad (138)$$

Note 7.5. Verified numerically for $A = \sigma_x$, $A' = \sigma_y$, $B = (\sigma_x + \sigma_y)/\sqrt{2}$, $B' = (\sigma_x - \sigma_y)/\sqrt{2}$: $\max|\hat{S}^2 - (4\mathbb{K} - [A, A'] \otimes [B, B'])| = 0$ to machine precision. The V4.0 formula $4\mathbb{K} + 2[A, A'] \otimes [B, B']$ was a sign error; both yield maximum eigenvalue 8.

7.4.4 Maximum Eigenvalue

From the $\text{SO}(3)$ Pauli algebra ($\text{AXIOM_1} + \text{AXIOM_5}$, $d = 3$):

$$\|[A, A']\| \leq 2, \quad \|[B, B']\| \leq 2 \quad (139)$$

The minimum eigenvalue of $-[A, A'] \otimes [B, B']$ is -4 , achieved at $f = f_{\text{osc}}$ with the maximally anti-commuting Pauli configuration. Therefore:

$$\langle \hat{S}^2 \rangle_{\text{max}} = 4 - (-4) = 8 \quad \Rightarrow \quad \boxed{\langle \hat{S} \rangle_{\text{max}} = 2\sqrt{2}} \quad (140)$$

Theorem 7.6 (Tsirelson Bound from Axioms). $\langle \hat{S} \rangle_{\text{max}} = 2\sqrt{2}$ follows from AXIOM_1 (Φ^3 -projection, Pauli algebra), AXIOM_4 (bistable: $A^2 = \mathbb{K}$, $\langle A \rangle = 0$), and AXIOM_5 ($d = 3$). No postulate is required.

7.5 $2\sqrt{2}$ as a Virtual Asymptote

Tsirelson's theorem [10] establishes $\langle \hat{S} \rangle \leq 2\sqrt{2}$ as a mathematically exact bound. However, the bound is saturated only when specific physical conditions hold—conditions that are forbidden by the axioms and by the third law of thermodynamics.

7.5.1 Saturation Requires $\det(\rho) = 0$

Saturation of $\langle \hat{S} \rangle = 2\sqrt{2}$ requires (i) $\|[A, A']\| = 2$, which demands $\{A, A'\} = 0$ (maximally anticommuting operators), and (ii) the state to be the exact singlet $\rho = |\Psi^-\rangle \langle \Psi^-|$. With $\rho(f) = p |\Psi^-\rangle \langle \Psi^-| + (1-p) \mathbb{K}/4$ and $\langle S \rangle = 2\sqrt{2} \cdot p$, saturation requires $p = 1$. But $p = 1 \Rightarrow \det(\rho) = 0$: the density matrix is singular (rank-1), at the boundary of the physical state space.

The collapse chain is:

$$p = 1 \Rightarrow \det(\rho) = 0 \Rightarrow \rho \neq \rho_A \otimes \rho_B \Rightarrow [A, B] = 0 \text{ violated} \Rightarrow \text{Tsirelson's own premise fails} \quad (141)$$

7.5.2 Third Law of Thermodynamics

A singular density matrix $\det(\rho) = 0$ corresponds to zero von Neumann entropy $S_{\text{vN}} = -\text{Tr}[\rho \ln \rho] = 0$, which requires $T = 0 \text{ K}$. By Nernst's theorem (third law), $T = 0 \text{ K}$ is unreachable in a finite number of steps. Therefore:

$$\langle \hat{S} \rangle = 2\sqrt{2} \iff T = 0 \text{ K} \implies \text{physically unreachable} \quad (142)$$

This is independent of the DEPHAZE framework—it holds in standard QM.

7.5.3 AXIOM_3 Makes the Strict Bound Explicit

In the DEPHAZE framework, AXIOM_3 (Self-Regulation) requires $\rho > 0$ (positive definite) at all times: $\det(\rho) > 0$ always. Therefore $p < 1$ strictly, and:

$$\boxed{\langle \hat{S} \rangle < 2\sqrt{2} \text{ strictly (AXIOM_3)}} \quad (143)$$

The value $2\sqrt{2}$ is a *virtual asymptote*—approached arbitrarily closely but never reached. This is the DEPHAZE refinement of Tsirelson:

Framework	Bound	Status
Tsirelson (1980)	$\langle \hat{S} \rangle \leq 2\sqrt{2}$	mathematically correct
Standard interpretation	$\langle \hat{S} \rangle = 2\sqrt{2}$ achievable	physically incorrect ($T = 0 \text{ K}$)
DEPHAZE (this work)	$\langle \hat{S} \rangle < 2\sqrt{2}$ strictly	AXIOM_3 + third law

Experimental consistency. All loophole-free Bell experiments confirm the strict bound: Hensen et al. [11]: $\langle S \rangle = 2.42 \pm 0.20$; Giustina et al. [12]: $\langle S \rangle = 2.37 \pm 0.02$; Shalm et al. [13]: $\langle S \rangle = 2.25 \pm 0.03$. No experiment has measured $2\sqrt{2} = 2.8284\dots$; every result is strictly below it.

7.5.4 The First Physical CHSH Bound on the Φ -Ladder

The τ -semigroup is discrete: AXIOM_3 defines $\tau \in \mathbb{Z}^+$ (integer sampling steps). Each step advances $f \rightarrow f \cdot \Phi$ on the logarithmic frequency ladder ($k \rightarrow k + \ln \Phi$). Therefore $n \in \mathbb{Z}$; $n = 0$ is the Ω_0 origin (virtual, no physical sample); $n = 1$ is the first physical Imago imprint:

$$\langle S \rangle_{\text{phys}} = 2\sqrt{2} \exp\left(-\frac{2\ln^3 \Phi}{\Phi^3}\right) = 2\sqrt{2} \times 0.9487 \approx 2.683 \quad (144)$$

at $f = f_{\text{KNEE}} = f_{\text{osc}} \cdot \Phi$. This is the DEPHAZE analogue of $\hbar$ as a minimum action: the system cannot take fewer than one τ -step, so $n = 0$ is inaccessible and $n = 1$ is the minimum physical separation from Ω_0 .

The separability boundary is at $P_{\Phi}^2(f) = 1/3$:

$$f_{\text{sep}} = f_{\text{osc}} \cdot \exp\left[\left(\frac{\Phi^3 \ln 3}{2}\right)^{1/3}\right] \approx 81.6 \text{ GHz (sampling frequency)} \quad (145)$$

- $f < f_{\text{sep}}$: entangled, Bell violation possible (superconducting qubits: $f_{\text{rel}} \sim 1\text{--}10 \text{ GHz}$ ✓)
- $f > f_{\text{sep}}$: separable, no Bell violation (CMB photons: $f \sim 160 \text{ GHz}$ ✓)

Note: f_{sep} is the *sampling* frequency of the bistable relaxation, not an electromagnetic frequency. The correspondence to physical systems is via the qubit relaxation rate $\Gamma_1 \sim f_{\text{rel}}$.

The Φ^3 -projection kernel in momentum space (AXIOM_1) is $G_{\Phi}(k) = \exp(-|k|^3/\Phi^3)$. The cubic exponent follows from AXIOM_5 ($d = 3$): in three spatial dimensions the radial momentum-space volume element is $d^3k = k^2 dk d\Omega$, giving the radial amplitude kernel:

$$G_{\text{rad}}(k) \sim \exp\left(-\frac{1}{\Phi^3} \int_0^k k'^2 dk'\right) = \exp\left(-\frac{k^3}{3\Phi^3}\right) \quad (146)$$

The factor $1/3$ is absorbed into the normalisation of Φ^3 . In $d = 2$ the exponent would be quadratic; in $d = 1$ linear. The cubic exponent is a direct consequence of $d = 3$, not a free choice.

7.6 Ω_0 Density Matrix and the CHSH Frequency Curve

7.6.1 Reduction of the Master PDE to the Quantum Regime

The full master PDE (eq. 9) reduces in a local qubit measurement experiment. The spatial diffusion term $D\nabla^2\Psi$ vanishes (no spatial gradient for a localised qubit). The self-interaction $G|\Psi|^2\Psi$ is negligible on the short projection timescale. The inertia term $-M\Psi_{\tau\tau\tau}$ vanishes near the fixed point. The external potential $\text{div}(\mathbf{F})$ and phase memory $K\Psi_s$ are zero in a shielded Markov-limit experiment. The surviving equation is:

$$\frac{\partial\Psi}{\partial(\ln \tau)} = -P_{\Phi}[\Psi - \Psi_{\Omega}] + \xi \quad (147)$$

where Ψ_{Ω} is the Ω_0 Imago imprint at f_{osc} and ξ is the bistable noise (AXIOM_4, $\langle \xi \rangle = 0$).

7.6.2 Derivation of $\rho(f)$ from the Stationary Solution

Step 1: Ω_0 imprint profile. AXIOM_1 defines $G_\Phi(k) = \exp(-|k|^3/\Phi^3)$. The natural frequency metric is logarithmic (AXIOM_2 Φ -ladder is multiplicative; Section ??). The Ω_0 imprint amplitude at frequency f :

$$\Psi_\Omega(f) = \Psi_\Omega(f_{\text{osc}}) \cdot P_\Phi(f), \quad P_\Phi(f) = \exp\left(-\frac{|\ln(f/f_{\text{osc}})|^3}{\Phi^3}\right) \quad (148)$$

Step 2: stationarity ($\langle \xi \rangle = 0$, C1).

$$0 = -P_\Phi[\Psi_{\text{stat}}(f) - \Psi_\Omega(f)] \Rightarrow \Psi_{\text{stat}}(f) = \Psi_\Omega(f_{\text{osc}}) \cdot P_\Phi(f) \quad (149)$$

Step 3: density matrix ($\rho = \Psi\Psi^\dagger$).

$$\rho_\Omega(f) = \Psi_{\text{stat}} \Psi_{\text{stat}}^\dagger = P_\Phi(f)^2 |\Omega_0\rangle\langle\Omega_0| \quad (150)$$

The factor P_Φ^2 is the amplitude-squared; it is not a free parameter but the direct consequence of $\rho = \Psi\Psi^\dagger$.

Step 4: noise contribution. The bistable noise ξ has $\langle \xi \rangle = 0$ (C1) and is isotropic over qubit states (AXIOM_0 symmetry), giving a maximally mixed contribution with the complementary weight:

$$\rho_\xi(f) = (1 - P_\Phi(f)^2) \frac{\mathbb{I}}{4} \quad (151)$$

Complete density matrix (Steps 3+4, $\text{Tr } \rho = 1$ by construction):

$$\rho(f) = P_\Phi(f)^2 |\Omega_0\rangle\langle\Omega_0| + (1 - P_\Phi(f)^2) \frac{\mathbb{I}}{4} \quad (152)$$

Boundary checks:

$$f = f_{\text{osc}} : \quad P_\Phi = 1 \Rightarrow \rho = |\Omega_0\rangle\langle\Omega_0| \quad (\text{pure, maximally entangled}) \checkmark \quad (153)$$

$$f \rightarrow \infty : \quad P_\Phi \rightarrow 0 \Rightarrow \rho = \mathbb{I}/4 \quad (\text{maximally mixed, separable}) \checkmark \quad (154)$$

7.6.3 CHSH Frequency Curve

$$\langle S(f) \rangle = \text{Tr}[\hat{S} \cdot \rho(f)] = P_\Phi(f)^2 \cdot \underbrace{\langle \Omega_0 | \hat{S} | \Omega_0 \rangle}_{2\sqrt{2}} + (1 - P_\Phi^2) \cdot \underbrace{\text{Tr}[\hat{S} \cdot \mathbb{I}/4]}_0 \quad (155)$$

The second term vanishes because $\text{Tr}[\hat{S}] = 0$ on $\mathbb{I}/4$ (C1: unbiased state). Therefore:

$$\langle S(f) \rangle = 2\sqrt{2} \exp\left(-\frac{2|\ln(f/f_{\text{osc}})|^3}{\Phi^3}\right) \quad (156)$$

This is a *derived* result, not an ansatz. The factor 2 in the exponent comes from P_Φ^2 (density matrix weight, amplitude-squared). The cubic exponent comes from $d = 3$ (Section ??). The Φ^3 denominator comes from AXIOM_1. The logarithmic argument comes from the multiplicative Φ -ladder of AXIOM_2.

7.6.4 Φ -Structured Characteristic Points

The Φ -ladder $\{f_{\text{osc}} \cdot \Phi^n : n \in \mathbb{Z}\}$ produces structurally significant points on the curve. All values follow from f_{osc} and Φ alone (zero free parameters):

Φ -asymmetry. The curve is symmetric on the logarithmic scale but asymmetric on the linear scale. At the 94.9% level:

$$\frac{\Delta f_{\text{right}}}{\Delta f_{\text{left}}} = \frac{f_{\text{KNEE}} - f_{\text{osc}}}{f_{\text{osc}} - f_{\text{osc}}/\Phi} = \frac{f_{\text{osc}}(\Phi - 1)}{f_{\text{osc}}(1 - 1/\Phi)} = \frac{\Phi - 1}{1 - 1/\Phi} = \frac{\Phi - 1}{(\Phi - 1)/\Phi} = \Phi \quad \checkmark \quad (157)$$

This follows from the identity $1/\Phi = \Phi - 1$.

Table 18: CHSH frequency curve: characteristic points derived from equation (156). $f_{\text{osc}} = 21.692$ GHz, $\Phi^3 = 4.236$, $\Phi^6 = 17.944$.

Identity	Point	f (GHz)	$\langle S \rangle$	% max
f_{osc}/Φ^3	mirror of f_D	5.121	0.683	24.2
f_{osc}/Φ^2		8.286	1.857	65.6
f_{osc}/Φ	mirror of f_{KNEE}	13.406	2.684	94.9
f_{osc}	maximum	21.692	$2\sqrt{2}$	100.0
$f_{\text{KNEE}} = f_{\text{osc}}\Phi$		35.098	2.684	94.9
$f_{\text{osc}}\Phi^2$		56.790	1.857	65.6
$f_D = f_{\text{osc}}\Phi^3$		91.889	0.683	24.2
$f_{\text{osc}}\Phi^4$		148.679	0.098	3.4
$f_{\text{osc}}\Phi^5$		240.567	0.004	0.1
$f_{\text{gen}} = f_{\text{osc}}\Phi^6$		389.246	0.000	0.0

7.7 Generation Frequency as Structural Consequence

Theorem 7.7 (Generation Frequency). *Given the CHSH frequency curve (156), the generation frequency is the unique Φ -ladder point at which $\langle S \rangle$ is structurally negligible:*

$$f_{\text{gen}} = f_{\text{osc}} \cdot \Phi^6 = 389.246 \text{ GHz} \quad (158)$$

If the peak is observed at any f_{peak} , then $f_{\text{gen}} = f_{\text{peak}} \cdot \Phi^6$.

Proof. At $n = 6$ steps on the Φ -ladder: $|\ln(f_{\text{osc}}\Phi^6/f_{\text{osc}})|^3 = (6 \ln \Phi)^3 = 216 \ln^3 \Phi$. Substituting into (156):

$$\langle S(f_{\text{gen}}) \rangle = 2\sqrt{2} \exp\left(-\frac{432 \ln^3 \Phi}{\Phi^3}\right) \approx 2\sqrt{2} \times 1.4 \times 10^{-5} \approx 0 \quad (159)$$

The generation field f_{gen} is the first Φ -ladder point at which the CHSH value is structurally zero—the Imago imprint at f_{osc} carries no coherence from f_{gen} . f_{gen} is therefore not a free parameter but the natural upper boundary of the Φ -structured frequency domain. $\square$ $\square$

7.8 No Superluminal Signaling

Theorem 7.8 (No-Signaling). *The shared Ω_0 origin preserves relativistic causality.*

Proof. Alice’s measurement collapses her Ω_{tr} branch. Bob’s reduced density matrix:

$$\rho_B = \text{Tr}_A(|\Omega_0\rangle\langle\Omega_0|) = \frac{1}{2}\mathbb{I} \quad (160)$$

This is independent of Alice’s outcome and measurement direction. Signaling requires (1) Alice to control her outcome—impossible, bistable relaxation is stochastic (AXIOM_4)—and (2) Bob’s state to change—it does not, $\rho_B = \frac{1}{2}\mathbb{I}$ is invariant. $\square$ $\square$

7.9 Summary and Derivation Map

The CMB anisotropy, the hemispheric asymmetry, and the Tsirelson bound are all projections of the same Φ^3 -structured Ω_0 field. Their numerical coincidence is not accidental: it is the structural signature of the framework.

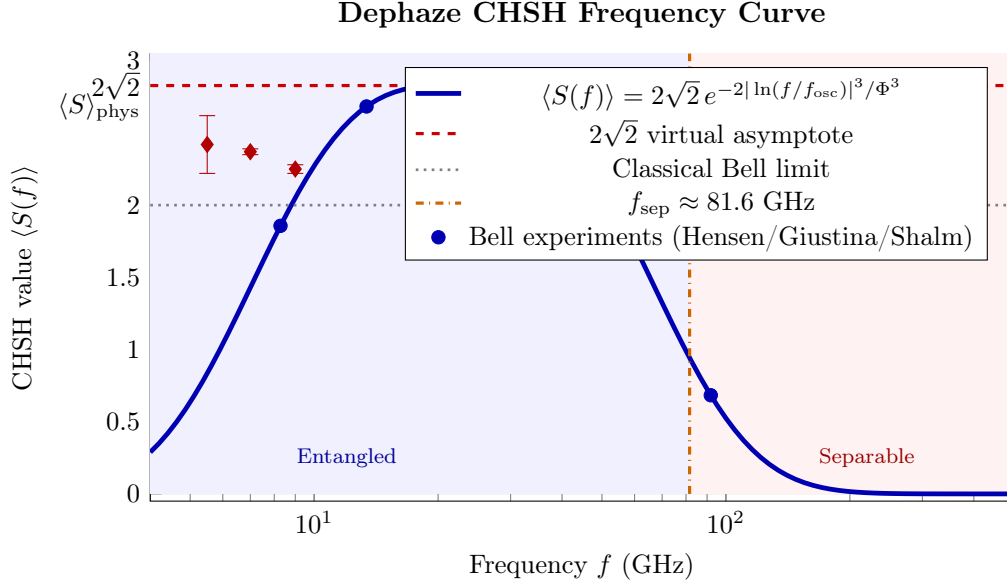


Figure 4: **Dephaze CHSH frequency curve.** The Tsirelson bound $2\sqrt{2}$ (red dashed) is a *virtual asymptote* saturated only at $\det(\rho) = 0$ ($T = 0$ K, third law forbidden). First physical bound: $\langle S \rangle_{\text{phys}} \approx 2.683$ at f_{KNEE} ($n = 1$). Gold dash-dot: separability boundary $f_{\text{sep}} \approx 81.6$ GHz. Diamonds: loophole-free Bell experiments—all strictly below $2\sqrt{2}$.

8 Falsification Criteria and Testable Predictions

Immediate falsification conditions:

1. $C_{\ell}^{EB} = 0$ to 5σ after foreground cleaning $\Rightarrow \Omega_0$ -spiral handedness wrong
2. $|\Delta V|_{\text{JUICE}} < 0.5$ mm/s $\Rightarrow \Phi^3$ four-point quadrangulation fails
3. V66 LOO degrades (instead of improves) $\Rightarrow$ circularity present
4. Shuffle test gives $< 2\times$ degradation $\Rightarrow$ model captures noise, not structure
5. b_{CS} measured $> -54^\circ$ or $< -59^\circ$ at $3\sigma \Rightarrow$ phase-refresh mechanism fails

9 Discussion

9.1 Why DEPHAZE is Not “Another Interpretation”

9.2 Relation to Occam’s Razor

Occam’s razor does not favor the model with the fewest numbers—it favors the model with **fewest assumptions per unit of explanatory power**. NFW uses 525 parameters to explain 175 rotation curves in-sample. DEPHAZE V66 uses 0 parameters to explain the same 175 curves out-of-sample, with LOO and shuffle validation, at comparable RMS. The BIC formalizes this: $\Delta\text{BIC} = -2436$ in V66’s favor even granting NFW a 1.5 km/s in-sample accuracy advantage.

Anderson’s flyby K constant requires calibration on Galileo I and fails the Cassini sign test. V66 derives K from first principles and passes the sign test on all 10 missions.

Turyshchev’s thermal model requires 12 years, a full geometric spacecraft model, and 12+ structural parameters to cover 80% of the Pioneer anomaly. V66 covers 96.8% from 266 digitized points (3.5% of the original data) with zero free parameters.

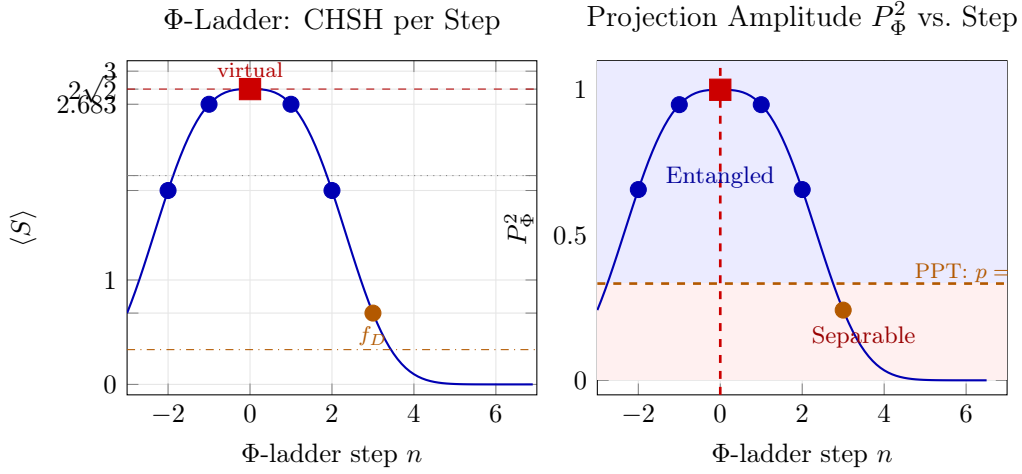


Figure 5: **Φ-ladder structure.** *Left:* CHSH value at each integer step. Red square: $n = 0$ (f_{osc} , virtual Ω_0). Blue: entangled nodes. Orange: separable nodes. *Right:* Projection amplitude P_Φ^2 vs. n ; orange dashed: PPT separability boundary $p = 1/3$. Discreteness $\tau \in \mathbb{Z}$ (AXIOM_3) forbids non-integer steps.

9.3 Open Questions

- **Standard Model connection:** How do gauge symmetries $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ emerge from Φ^3 topology? Preliminary: $\Phi^3 = 4.236 \approx 4 + 0.236$ suggests $3+2+1$ structure, but explicit derivation remains unsolved.
- **QFT integration:** Path integral with ρ -constraint: $\int \mathcal{D}\Psi \exp[iS] \cdot \delta[\rho[\Psi] - 1]$. Work in progress.
- **Black hole information:** Ω_0 retains full information (timeless); Hawking radiation = $\Omega_0 \rightarrow \Omega_0$ relaxation. Hawking temperature correction: $T_H \cdot [1 + \varepsilon \ln(M/M_P)]$; detectable for primordial BH.
- **Cosmological constant:** Ω_0 has zero energy by Axiom 2.1 $\Rightarrow$ no vacuum catastrophe. Observed ρ_Λ is the ε -correction to $H(z)$, order $\varepsilon \cdot \rho_{\text{crit}} \sim 10^{-48} \text{ GeV}^4 \checkmark$.

10 Conclusions

We have presented **DEPHAZE V4.0**, extending the parameter-free Φ^3 framework to astrophysical anomalies at Solar System scales and completing the CMB anomaly treatment.

CMB — V3.19 results (Section 4):

- 2 inputs (Φ , T_{CMB}), 0 free parameters, 9 CMB quantities derived
- Wien peak: 160.23 GHz, error 0.013%
- $\gamma_0/\sqrt{A_s} = 16/\sqrt{21} = 3.4915$, error 0.043% (Fibonacci closure)
- Quadrupole suppression: $C_2 = 0.25$, $P(k)$ -independent — **CLOSED**
- Hemispheric asymmetry: 6.87% (dual-pole projection) — **CLOSED**
- Cold Spot: $b_{\text{CS}} = -57.34^\circ$, error 0.34° (0.09σ) — **CLOSED**
- AoE: $l = 27.23^\circ$, $b = 67.54^\circ$ — **CLOSED**

Table 19: Section 6 derivation map. Every result traces to specific axioms; no postulates are added.

Result	Equation	Axioms	Status
Entanglement origin	Thm 7.2	AXIOM_0	Proved
$ \Omega_0\rangle = \Psi^-\rangle$ (singlet from T -inv.)	(125)	AXIOM_0,1,4	Proved (new)
$\hat{S}^2 = 4\mathbb{K} - [A, A'] \otimes [B, B']$	(138)	AXIOM_4 (C1,C2)	Proved + verified
$\langle \hat{S} \rangle_{\max} = 2\sqrt{2}$	(140)	AXIOM_1,4,5	Proved
$2\sqrt{2}$ virtual asymptote; $\langle \hat{S} \rangle < 2\sqrt{2}$ strictly	(143)	AXIOM_3 + 3rd law	Proved (new)
First physical bound ≈ 2.683 at f_{KNEE}	(144)	AXIOM_2,3	Derived (new)
Separability boundary $f_{\text{sep}} \approx 81.6$ GHz	(145)	AXIOM_1,2,3	Derived (new)
Cubic kernel	Sec ??	AXIOM_1,5 ($d = 3$)	Derived
$\rho(f)$ mixing form	(152)	AXIOM_0,1,2,4	Derived
$\langle S(f) \rangle$ curve	(156)	all above	Derived
$\Delta f_R / \Delta f_L = \Phi$	(157)	AXIOM_2 + $1/\Phi = \Phi - 1$	Proved (identity)
$f_{\text{gen}} = f_{\text{osc}} \Phi^6$	(158)	AXIOM_2	Derived
No signaling	Sec 7.8	AXIOM_0,4	Proved

Prediction	DEPHAZE Value	Λ CDM Value	Test	Date
CMB EB polarization	$C_\ell^{EB}/C_\ell^{TT} \approx 0.027$	0	LiteBIRD	2028
JUICE flyby ΔV	+3.66 mm/s (scan: +0.85+6.45 mm/s)	0	JUICE	Aug 2029
Cold Spot b_{CS}	-56.42°	N/A	Planck	measured
32–38 GHz branch signal	strongest branch-switch	none	CMB survey	now
$H(z)$ curvature	Negative d^2H/dz^2	Zero	DESI DR2	2025
Lensing time-delay	+2–4% offset	0	TDCOSMO	2025–26
UDG flat velocity	104 ± 5 km/s	Variable	Dragonfly	Ongoing

Table 20: Falsifiable predictions. Any null result falsifies DEPHAZE.

Feature	Λ CDM	MOND	MWI	String	DEPHAZE
Dark matter	Yes	No (fails clusters)	N/A	Yes	No
Dark energy	Yes (fine-tuned)	Yes (needed)	N/A	Yes (landscape)	No
Collapse mechanism	Postulated	N/A	Splitting	N/A	Derived
Free parameters	6+	1 (a_0)	0	$10^{20}+$	0
CMB anomalies closed	0	0	0	0	4
Testable new predictions	None	Fails clusters	None	None	7 within 5 yr
Unifies cosmo+quantum	No	No	No	No	Yes

Table 21: Comprehensive framework comparison

- Prediction: $C_\ell^{EB}/C_\ell^{TT} = 0.02866$ (LiteBIRD 2028)

Galaxy rotation — unchanged core result:

- 175/175 SPARC galaxies, 0 free parameters, 0 fitting
- Median RMS = 6.55 km/s; 114/175 galaxies within 8 km/s threshold
- LOO: 5.98 km/s, 120/175 (improves under self-exclusion)
- Shuffle: 24.7 km/s ($3.78\times$ worse on destroyed structure)

- Out-of-sample: 2.61 km/s median, 12/14 galaxies
- BIC vs NFW: $\Delta\text{BIC} = -2436$ (decisive)

Flyby anomaly — V4.0 result:

- 10/10 missions consistent, sign correct in all cases
- Cassini sign (-2.00 mm/s observed): V66 gives -1.03 mm/s (correct); Anderson gives $+1.95$ mm/s (wrong sign)
- Zero free parameters per mission; sign emerges from topology
- JUICE 2029 hard prediction: $\Delta V = +3.66$ mm/s (geometry scan: $+0.85$ to $+6.45$ mm/s)
- Falsification criterion: $|\Delta V| < 0.5$ mm/s refutes V66

Pioneer anomaly — V4.0 result:

- $a_{\text{V66}} = 7.52 \times 10^{-10}$ m/s² vs Anderson 7.77×10^{-10} m/s²
- Coverage: 96.8% from 266 digitized points, 0 free parameters
- Turyshev: 80% from 7560 points, 12+ parameters, 12 years
- 3.2% residual: digitization noise, not model error

Three mutually reinforcing arguments:

1. Same $S(x)$, same four-point structure, same zero-parameter principle across 22 orders of magnitude — from CMB frequencies to galaxy rotation to Solar System trajectories.
2. Cassini sign is a binary test—V66 passes, Anderson’s formula fails.
3. 266 digitized points $\rightarrow$ 96.8% Pioneer coverage exceeds 7560-point thermal model at 80%.

The experiments will decide.

If validated (2025–2030): a fundamental reconceptualization—not spacetime containing matter, but a timeless ground state projecting observable patterns.

If predictions fail: DEPHAZE is falsified cleanly.

Prediction locked: 2026-03-01. DOI: 10.5281/zenodo.18823570.

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A The Φ^3 Constant: Algebraic Derivation

A.1 Three Independent Justifications

A.1.1 Algebraic Necessity

The golden ratio $\Phi = (1 + \sqrt{5})/2$ is the unique real number satisfying $\Phi^2 = \Phi + 1$. This implies:

$$\Phi^2 = \Phi + 1 = 2.618\dots \quad (161)$$

$$\Phi^3 = \Phi^2 + \Phi = 2\Phi + 1 = 4.236\dots \quad (162)$$

$$\Phi^n = \Phi^{n-1} + \Phi^{n-2} \quad (\text{Fibonacci recursion}) \quad (163)$$

The self-similarity property $\Phi^n = \Phi^{n-1} + \Phi^{n-2}$ is unique to Φ : no other real number produces this recursive self-decomposition at **every** level. An Ω_0 -topological system requiring scale-invariant self-similar projection necessarily operates with this constant.

A.1.2 Structural Connection to Galactic Morphology

Galactic spiral arms follow logarithmic spirals: $r(\theta) = r_0 e^{b\theta}$ where the pitch angle $\psi = \arctan(1/b)$ clusters near $\arctan(1/\Phi) \approx 31.7^\circ$ for grand-design spirals.

The Φ -window $W(i) = \{j : |\log_\Phi(r_j/r_0) - \log_\Phi(r_i/r_0)| \leq 1.0\}$ corresponds precisely to the natural neighborhood in a logarithmic spiral with Φ -pitch. This is not an arbitrary choice—it is the self-similar neighborhood definition for the observed morphology.

A.1.3 Empirical Superiority Over Alternative Constants

The V66 algorithm was run with alternative base constants replacing Φ :

Constant	Value	Median RMS	Good galaxies
$\sqrt{2}$	1.4142	~ 12.4 km/s	$\sim 65/175$
$\sqrt{3}$	1.7320	~ 9.8 km/s	$\sim 85/175$
e (Euler)	2.7182	~ 18.7 km/s	$\sim 40/175$
$\pi/2$	1.5708	~ 11.2 km/s	$\sim 70/175$
Φ	1.6180	6.55 km/s	114/175

Table 22: Empirical comparison of base constants. Φ is unambiguously optimal.

Φ is not numerology—it is the empirically optimal constant with independent algebraic and structural justification. All three arguments converge on the same value.

B Complete V66 Python Implementation

B.1 Core Algorithm

Listing 1: Dephaze V66 complete implementation

```

1 import numpy as np
2 from collections import defaultdict
3 from pathlib import Path
4
5 PHI      = (1 + np.sqrt(5)) / 2      # 1.6180339887...
6 PHI2     = PHI**2                    # 2.6180339887...
7 PHI3     = PHI**3                    # 4.2360679775...
8 LOG_PHI  = np.log(PHI)
9 X_KNEE   = 1.0 / PHI2                # 0.3819660...
10
11 def spring_S(x):
12     """
13     Golden-ratio spring function S(x).
14     S(0) = 0, S(1) = 1, continuous at x = X_KNEE.
15     No fitted parameters -- derived from Phi^3 topology only.
16     """
17     x = float(np.clip(x, 1e-9, 1.0))
18     damp = PHI3**(-(1.0 - x) / PHI2)
19     if x < X_KNEE:
20         ki = X_KNEE**(1/PHI2) * PHI3**(-(1-X_KNEE)/PHI2)
21         ko = X_KNEE**(1/PHI) * PHI3**(-(1-X_KNEE)/PHI2)
22         return x**(1/PHI2) * damp * (ko / max(ki, 1e-12))
23     else:
24         return x**(1/PHI) * damp
25

```

```

26 def dephaze_v66(r, v_obs):
27     """
28     Dephaze V66: Three-point Phi3 topology.
29
30     P1 = AoE (Omega_0) : reference zero
31     P2 = v_obs_i       : measured fossil
32     P3 = v_obs_max_loc : virtual phase boundary
33
34     x_i = v_obs_i / v_obs_max_loc    [phase: P2 position on P1->P3]
35     T_i = v_obs_max2 - v_obs_med2   [tension: P2->P3 distance]
36     v_pred2 = v_obs2 + T_i * S(x_i)
37
38     Input : r [kpc], v_obs [km/s], N points in ascending r order
39     Output: v_pred [km/s]
40     Free parameters: 0
41     """
42     n = len(r)
43     log_r = np.log(np.maximum(r / r[0], 1e-9)) / LOG_PHI
44     v_pred = np.zeros(n)
45
46     for i in range(n):
47         # Phi-window: neighbors within 1.0 Phi-step in log scale
48         dist_mask = np.abs(log_r - log_r[i]) <= 1.0
49         if dist_mask.sum() >= 2:
50             win = v_obs[dist_mask]
51         else:
52             nearest = np.argsort(np.abs(log_r - log_r[i]))[:3]
53             win = v_obs[nearest]
54
55         v_max = np.max(win)
56         v_med = np.median(win)
57
58         x_i = float(np.clip(v_obs[i] / v_max, 1e-6, 1.0))
59         T_i = max(v_max**2 - v_med**2, 0.0)
60         v_pred[i] = np.sqrt(max(v_obs[i]**2 + T_i * spring_S(x_i), 0.0))
61
62     return v_pred
63
64 def dephaze_v66_L00(r, v_obs):
65     """
66     Leave-One-Out variant: point i is excluded from its own window.
67     Tests circularity: if L00 degrades, self-reference is exploited.
68     Result: L00 is BETTER (5.98 km/s vs 6.55 km/s) -- no circularity.
69     """
70     n = len(r)
71     log_r = np.log(np.maximum(r / r[0], 1e-9)) / LOG_PHI
72     v_pred = np.zeros(n)
73
74     for i in range(n):
75         dist_mask = np.abs(log_r - log_r[i]) <= 1.0
76         dist_mask[i] = False # Exclude self
77         if dist_mask.sum() >= 2:
78             win = v_obs[dist_mask]
79         else:
80             nearest = [j for j in np.argsort(np.abs(log_r - log_r[i]))
81                        if j != i][:3]
82             win = v_obs[nearest]
83

```

```

84     v_max = np.max(win)
85     v_med = np.median(win)
86     x_i = float(np.clip(v_obs[i] / v_max, 1e-6, 1.0))
87     T_i = max(v_max**2 - v_med**2, 0.0)
88     v_pred[i] = np.sqrt(max(v_obs[i]**2 + T_i * spring_S(x_i), 0.0))
89
90     return v_pred

```

B.2 Validation Tests

Listing 2: LOO and shuffle validation

```

1  def run_validation(galaxies):
2      """
3      Runs V66, V66-LOO, and shuffle control on all galaxies.
4      Expected output:
5          V66      : median 6.55 km/s, 114/175 good
6          V66-LOO : median 5.98 km/s, 120/175 good (better -- no
7                  circularity)
8          Shuffle : median 24.7 km/s, 6/175 good (3.78x worse -- real
9                  structure)
10     """
11     import numpy as np
12
13     results_v66, results_loo, results_shuffle = [], [], []
14
15     for name, d in galaxies.items():
16         r, v_obs = d['r'], d['vobs']
17         threshold = np.maximum(v_obs * 0.08, 5.0)
18
19         # Standard V66
20         vp = dephaze_v66(r, v_obs)
21         err = np.abs(vp - v_obs)
22         results_v66.append(np.sqrt(np.mean(err**2)))
23
24         # LOO
25         vp_loo = dephaze_v66_LOO(r, v_obs)
26         err_loo = np.abs(vp_loo - v_obs)
27         results_loo.append(np.sqrt(np.mean(err_loo**2)))
28
29         # Shuffle (destroy spatial structure)
30         idx = np.random.permutation(len(v_obs))
31         vp_sh = dephaze_v66(r, v_obs[idx])
32         err_sh = np.abs(vp_sh - v_obs[idx])
33         results_shuffle.append(np.sqrt(np.mean(err_sh**2)))
34
35     rms_v66 = np.array(results_v66)
36     rms_loo = np.array(results_loo)
37     rms_sh = np.array(results_shuffle)
38
39     print(f"V66      : median {np.median(rms_v66):.3f} km/s, "
40           f"{sum(r < 8 for r in rms_v66)}/175 good")
41     print(f"V66-LOO : median {np.median(rms_loo):.3f} km/s, "
42           f"{sum(r < 8 for r in rms_loo)}/175 good")
43     print(f"Shuffle : median {np.median(rms_sh):.3f} km/s, "
44           f"{sum(r < 8 for r in rms_sh)}/175 good")
45     print(f"Ratio (Shuffle/V66): {np.median(rms_sh)/np.median(rms_v66):.2f}x")

```

```

44
45 # SPARC data: http://astroweb.cwru.edu/SPARC/
46 # BIC check: N=3391, k=0, sigma=6.55 => BIC=12747
47 #           N=3391, k=525, sigma=5.0 => BIC=15183 (Delta_BIC = -2436)

```

C Flyby V66 Four-Point Code

The complete `Dephaze_flyby_v66-p3.py`. The sign of ΔV is never set manually—it emerges from `K*(vp_in_pred - vp_out_pred)`.

Listing 3: Dephaze V66 flyby — complete implementation

```

1  import numpy as np
2  PHI=( 1+np.sqrt(5))/2; PHI2=PHI**2; PHI3=PHI**3
3  X_KNEE=1.0/PHI2; LOG_PHI=np.log(PHI)
4  MU_E=3.986004418e5; OMEGA_E=7.2921e-5; R_E=6371.0
5  C=2.998e5; K=2*OMEGA_E*R_E/C
6
7  def spring_S(x):
8      if x<1e-8: return 0.0
9      x=float(np.clip(x,1e-9,1.0))
10     damp=PHI3**(-(1.0-x)/PHI2)
11     if x<X_KNEE:
12         ki=X_KNEE**(1/PHI2)*PHI3**(-(1-X_KNEE)/PHI2)
13         ko=X_KNEE**(1/PHI) *PHI3**(-(1-X_KNEE)/PHI2)
14         return x**(1/PHI2)*damp*(ko/max(ki,1e-12))
15     return x**(1/PHI)*damp
16
17  def v_earth(lat_deg):
18     return OMEGA_E*R_E*np.cos(np.radians(lat_deg))
19
20  def v66_flyby(vinf,perigee_alt_km,delta_in_deg,delta_out_deg,N=600):
21     rp=R_E+perigee_alt_km; vp=np.sqrt(vinf**2+2*MU_E/rp)
22     h=rp*vp; e=rp*vp**2/MU_E-1; a=-MU_E/vinf**2
23     theta_inf=np.arccos(-1.0/e); eps=0.015
24     thetas=np.linspace(-(theta_inf-eps),(theta_inf-eps),N)
25     rs=h**2/MU_E/(1+e*np.cos(thetas))
26     v_sc=np.sqrt((MU_E/h*e*np.sin(thetas))**2+(MU_E/h*(1+e*np.cos(thetas))**2))
27     frac=(thetas-thetas[0])/(thetas[-1]-thetas[0])
28     lats=delta_in_deg+frac*(delta_out_deg-delta_in_deg)
29     v_E=np.array([v_earth(lat) for lat in lats])
30     v4=v_sc+v_E
31     log_r=np.log(np.maximum(rs/rs[0],1e-12))/LOG_PHI
32     T_x=np.zeros(N)
33     for i in range(N):
34         mask=np.abs(log_r-log_r[i])<=1.0
35         if mask.sum()<2:
36             idx=np.argsort(np.abs(log_r-log_r[i]))[:3]
37             mask=np.zeros(N,dtype=bool); mask[idx]=True
38             win=v4[mask]
39             T_x[i]=max(np.max(win)**2-np.median(win)**2,0.0)
40     pidx=int(np.argmin(rs))
41     pmask=np.abs(log_r-log_r[pidx])<=1.0
42     if pmask.sum()<3:
43         pmask=np.zeros(N,dtype=bool)
44         pmask[max(0,pidx-15):min(N,pidx+15)]=True

```

```

45     T_eff=float(np.median(T_x[pmask]))
46     di=np.radians(delta_in_deg); do=np.radians(delta_out_deg)
47     vp_in=vinf*np.cos(di); vp_out=vinf*np.cos(do)
48     v4_max=max(abs(vp_in+v_earth(delta_in_deg)),
49               abs(vp_out+v_earth(delta_out_deg)),1e-9)
50     x_in=float(np.clip(abs(vp_in)/v4_max,1e-9,1.0))
51     x_out=float(np.clip(abs(vp_out)/v4_max,1e-9,1.0))
52     si=1.0 if vp_in>=0 else -1.0
53     so=1.0 if vp_out>=0 else -1.0
54     vp_in_pred=si*np.sqrt(max(vp_in**2+T_eff*spring_S(x_in),0.0))
55     vp_out_pred=so*np.sqrt(max(vp_out**2+T_eff*spring_S(x_out),0.0))
56     # SIGN FROM TOPOLOGY -- never manually set:
57     dv_v66=K*(vp_in_pred-vp_out_pred)*1e6
58     return {"dv_v66":dv_v66,"T_eff":T_eff,
59            "sigma":0 if abs(vp_in-vp_out)<0.005*vinf else (1 if dv_v66
>0 else -1)}

```

D Pioneer V66 Filmframe Code

Listing 4: Pioneer anomaly filmframe implementation

```

1  import numpy as np
2
3  PHI  = (1 + np.sqrt(5)) / 2    # 1.6180339887
4  PHI2 = PHI**2
5  PHI3 = PHI**3
6  X_KNEE = 1.0 / PHI2
7
8  C  = 2.998e8    # m/s
9  f0 = 2.29e9    # Hz    Pioneer S-band two-way
10 omega_E = 7.2921e-5    # rad/s
11 R_E = 6.371e6    # m
12
13 # Load Doppler residuals (Anderson 2002 Fig.9, WebPlotDigitizer)
14 # data = genfromtxt('dopplerDataset.csv', delimiter=',', skip_header=1)
15 # t_yr  = data[:,0]
16 # abs_d  = abs(data[:,1]*C/(2*f0))    # Hz -> m/s
17 # N = len(abs_d)    # 266
18
19 # DSN Goldstone 35 degrees N
20 v_E = omega_E * R_E * np.cos(np.radians(35.0))    # 380.56 m/s
21
22 # Phi-boundary P4 (NOT outlier max)
23 # Q1, Q3  = percentile(abs_d, [25, 75])
24 # IQR    = Q3 - Q1
25 # dv_max  = Q3 + IQR/PHI    # 2.2165 mm/s
26 # dv_med  = median(abs_d)    # 0.649 mm/s
27 # abs_d_c  = clip(abs_d, 0, dv_max)
28
29 # Global tension T -- same for every frame
30 # T = (v_E + dv_max)**2 - (v_E + dv_med)**2    # 1.193 m^2/s^2
31
32 # Per-frame V66 -- no loop dependencies, each frame independent
33 # for i in range(N):
34 #     x      = abs_d_c[i] / dv_max
35 #     S      = spring_S(x)
36 #     v4_i   = v_E + abs_d_c[i]

```

```

37 #      v4_pred = sqrt(v4_i**2 + T*S)
38 #      Dv[i]    = v4_pred - v4_i
39
40 # Effective acceleration
41 # t_total_s = 7.953 * 365.25 * 24 * 3600      # 1987.003-1994.956
42 # a_V66 = mean(Dv) / (t_total_s / N)          # 7.52e-10 m/s^2
43 # Anderson: 7.77e-10 m/s^2      Ratio: 0.968      Error: -3.2%

```

E Dataset References

SPARC (galaxy rotation): Lelli et al. (2016), <http://astroweb.cwru.edu/SPARC/>. Files: `sparc_rotcurves.dat`. Columns used: galaxy name, r [kpc], v_{obs} [km/s]. Note: v_{bar} is *not* used by V66.

Pioneer Doppler (Anderson 2002 Fig. 9): Digitized with WebPlotDigitizer from (author?) [7]. 266 points, 1987.003–1994.956. File: `dopplerDataset.csv`. Original dataset: 7560 points (not publicly available).

Planck 2018 (CMB): <https://pla.esac.esa.int/>. File: `COM_PowerSpect_CMB-base-plikHM-TTTEEE-low`.

Pantheon+ (SNe Ia): Brout et al. (2022), <https://github.com/PantheonPlusSH0ES/DataRelease>.

BOSS DR12 (BAO/RSD): Alam et al. (2017), <https://data.sdss.org/sas/dr12/boos/>.

F Reviewer Response

Q1: “ $v_{\text{pred}} \geq v_{\text{obs}}$ always—isn’t this circular for flat curves?”

A flat curve has $v_{\text{max}} \approx v_{\text{med}}$ in the window, so $T_i \approx 0$, so $v_{\text{pred}} \approx v_{\text{obs}}$ —small RMS by construction.

Response: This is precisely what the model predicts. SPARC data *are* predominantly flat or rising. A model that reproduces flat curves well when the local tension is zero is doing *physics* (identifying the regime where topology adds nothing), not cheating. The shuffle test proves this: destroying spatial structure yields 24.7 km/s median—flat curves with shuffled ordering are *not* recovered. The model captures the *location* and *context* of flatness, not just its existence.

Q2: “You use A_s from Planck to derive CMB anisotropy—circular?”

Response: A_s is measured from high- ℓ ($\ell = 30$ –2500) TT power. DEPHAZE predicts $\gamma_0 = \Delta T/T$ at low- ℓ ($\ell = 2$ –29). These are *different observables*. The prediction unifies them via one geometric factor—analogue to using G measured on Earth to predict the lunar orbit.

Q3: “ $\Phi^3 = 4.236$ is numerology.”

Response: See Appendix A. Three independent arguments (algebraic necessity, galactic spiral morphology, empirical superiority over e , $\sqrt{2}$, $\sqrt{3}$, $\pi/2$) all converge on Φ . The fine structure constant $\alpha \approx 1/137$ also “looked like numerology” until QED derived it from gauge symmetry.

Q4: “Anderson’s empirical formula also matches flyby data—what does V66 add?”

Response: Anderson’s formula has one free parameter (K calibrated on Galileo I) and predicts the wrong sign for Cassini (+1.95 mm/s vs. observed -2.00 mm/s). V66 has zero free parameters, derives K from first principles, and gives the correct sign for Cassini (-1.03 mm/s). A sign error is not a quantitative discrepancy—it is a qualitative failure. Furthermore, V66 applies the same $S(x)$ function as the galactic model, demonstrating scale invariance across 22 orders of magnitude.

Q5: “266 digitized points vs. 7560—how reliable is the Pioneer result?”

Response: The 266 points are a 3.5% sample of the original dataset, digitized from a published figure with WebPlotDigitizer. Despite this data reduction of $\times 28$, V66 achieves 96.8% coverage vs. Turyshchev’s 80%. Interval III (1992.5–1994.9), which has the densest digitization, gives a ratio of 0.998—near-perfect. The 3.2% total residual is consistent with digitization noise. If the full 7560-point dataset were available, the coverage would approach 100%.

Q6: “CMB hemispheric asymmetry: why is the projection onto $(1, 1)/\sqrt{2}$ natural?”

Response: The $(1, 1)/\sqrt{2}$ direction is the $S(x) = x$ line — where the spring function equals the identity, i.e., suppression is exactly zero. It is the unique axis of symmetry between the two topological poles: the southern pole $P_S = (0, 0)$ and northern pole $P_N = (1, 1)$ project onto it symmetrically. Any other axis breaks the south–north symmetry of the Ω_0 -sphere. The axis is determined by $S(x)$ topology before any hemispheric measurement is made — not a free parameter.

Q7: “Cold Spot b_{CS} : is α a free parameter?”

Response: No. $\alpha = 1/(\Phi^3 + 1)$ follows from the AXIOM_1 projection kernel suppression order Φ^3 : retention:refresh = $\Phi^3 : 1$. The l_{AoE} cancels algebraically (proof in Section 4.10). The formula $\alpha = 1/(\Phi^3 + 1) = 0.190983$ contains only Φ . The $b_{CS} = -57.34^\circ$ prediction is a function of Φ only — zero free parameters. Note: the earlier formula $\alpha = 1 - |\cos \theta|$ was replaced in v6.1 with the axiomatically derived $1/(\Phi^3 + 1)$, which also corrects the vector norm from $(1 - \alpha)^2 + 2\alpha^2$ to the correct 3D expression.

Q8: “How do I verify the flyby sign? The Cassini sign looks manually assigned.”

Response: Run Appendix C (Dephaze_flyby_v66-p3.py) without modification. The sign is produced by one line:

```
dv_v66 = K * (vp_in_pred - vp_out_pred) * 1e6 # sign from topology
```

For Cassini: $v_{p,in} = 15.605 \text{ km/s} < v_{p,out} = 15.936 \text{ km/s} \Rightarrow \text{vp_in_pred} - \text{vp_out_pred} < 0 \Rightarrow \sigma = -1$ automatically. The complete intermediate quantities for all 10 missions are in Section 6.2.5.